Geoderma 228–229 (2014) 67–89

Contents lists available at ScienceDirect

Geoderma

journal homepage: www.elsevier.com/locate/geoderma

Accelerated soil formation due to paddy management on marshlands (Zhejiang Province, China)

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ARTICLE INFO

Article history:
Received 30 July 2012
Received in revised form 19 June 2013
Accepted 5 September 2013
Available online 3 October 2013

Keywords:
Phases of pedogenesis
Paddy soils
Soil organic matter
Soil mineralogy
Lipid biomarker
Decalcification

ABSTRACT

Inundation of paddy soils for submerged rice production strongly impacts soil formation. Here we used chronosequences with up to 2000 years of cultivation history to compare soil formation in non-inundated (non-paddy) cropping systems with the formation of soils used for paddy rice production. This approach allowed us to identify the influence of agricultural management at different stages of pedogenesis.

Soil samples were taken from two chronosequences derived from uniform parent material in the coastal region of the Zhejiang Province (P.R. China). One chronosequence consisted of paddy soils of different ages (50–2000 years), characterized by a yearly cropping sequence of rice cultivation under flooded conditions alternated with a non-inundated crop. The adjacent non-inundated (non-paddy) chronosequence was exclusively used for non-inundated crop production for 50–700 years. Lipid biomarkers revealed origin and homogeneity of the original coastal sediments and enabled the reconstruction of a consistent land use history for both chronosequences.

The chronological development of soil properties and horizons suggested that the formation of paddy soils can be subdivided into three phases. The initial phase of paddy soil development takes only a few decades and is dominated by desalinization and formation of a compacted plow pan, leading from Fluvisols to Anthraquic Cambisols. During the next centuries (second phase), the differentiation between paddy and non-paddy management becomes increasingly obvious in terms of accelerated carbonate losses and constantly increasing organic carbon concentrations in paddy topsoils. In the third stage of paddy soil development (≥700 years), a (trans-)formation and redistribution of oxides is accompanied by clearly visible hydromorphic patterns in paddy subsoils, thus promoting further development from Cambisols to Hydragric Anthrosols. To account for the underlying processes we suggest modifying the depth and mottling criteria for the definitions of anthraquic and hydragric soil horizons in the classification of the World Reference Base for Soil Resources. The non-paddy chronosequence was characterized by a low degree of soil development in which decalcification-related processes dominated throughout 700 years of soil formation. Hence, soil formation under paddy management was accelerated relative to that under dryland cropping, even though the 2000-year-old paddy soils lacked evidence of an advanced stage of silicate weathering, formation of pedogenic clay minerals, or clay migration.

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1. Introduction

Cultivated rice (*Oryza sativa*) is an annual grass mostly grown under flooded lowland conditions in bunded fields (paddies). The traditional way of growing paddy rice involves the tillage of flooded soil (puddling), followed by transplantation of rice plants and crop growth in a submerged environment from crop establishment until close to harvest. At the end of the growing season, paddy fields are drained about 2 or 3 weeks before harvest (Janssen et al., 2006). Worldwide, there are about 133 million ha of flooded lowland rice fields, of which some 79 million ha are equipped with irrigation facilities (http://www.knowledgebank.irri.org/rkb/watermanagement.html).

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0016-7061/$ – see front matter © 2013 Elsevier B.V. All rights reserved.
http://dx.doi.org/10.1016/j.geoderma.2013.09.005

Paddy soil formation is induced by puddling and a flooding and drainage regime leading to characteristic redoximorphic features. Long-term paddy rice cultivation results in a marked alteration of hydromorphic soil properties due to specific processes of paddy soil formation initiated by redox potential oscillations. Frequent inundation leads to anoxic conditions in the soil and reduced rates of organic matter decomposition (Sahrawat, 2004, 2005), while it intensifies mineral weathering (Nanzyo et al., 1999) and leaching processes (Gong, 1986; Ponnamperuma, 1972). Recurrent redox alternations enhance the changes and the release of structural Fe in clay minerals (Favre et al., 2002) and lead to a translocation of Fe in various directions (Gong, 1986; Ratering and Schnell, 2000). High concentrations of dissolved organic matter during flooding seasons enhance the changes as well as the release of structural Fe in clay minerals and support the formation of ferrihydrite (Kögel-Knabner et al., 2010; Rancourt et al., 2005). Frequent redox cycles may favor the formation of poorly crystalline hydrous Fe oxides, i.e., ferrihydrite. However, Thompson et al. (2006) observed a positive correlation between the number of redox cycles and increasing crystallinity of soil hydrous oxides in short-term batch experiments. This could also result in higher total Fe oxide contents in paddy soils compared to that in non-paddy soils (Cheng et al., 2009).

The formation of specific properties due to paddy soil management leads to Hydragric Anthrosols according to the World Reference Base for Soil Resources (WRB) (Driessen et al., 2001; Gong, 1983; IUSS Working Group WRB, 2007). Anthrosols developed under paddy cultivation are defined by the presence of an anthraquic horizon overlying a hydragric horizon, with a total minimum thickness of 50 cm. The anthraquic horizon, which develops first, is at least 20 cm thick and consists of a puddled layer overlying a plow pan. The puddled layer shows sorted soil aggregates, vesicular pores, and a specific Munsell color. The plow pan has a platy structure, a bulk density (BD) that is at least 20% higher than that of the puddled layer, and yellowish-brown, brown, or reddish-brown Fe–Mn mottles or coatings. With ongoing paddy soil management the hydragric subsoil horizon develops. This horizon is the result of wet cultivation and is at least 10 cm thick. It is characterized by the accumulation of Fe and Mn mobilized by the surface water that slowly penetrates the plow pan, resulting in Fe–Mn mottles or coatings. Therefore, this horizon is characterized by Fe or Mn coatings or concretions, by redox-depleted zones in macropores, or by higher dithionite–citrate extractable Fe or Mn compared with the surface horizon (IUSS Working Group WRB, 2007). The presence of the two diagnostic horizons is expressed by the Hydragric qualifier. If only the anthraquic horizon is developed, the Anthraquic qualifier is used. In the latter case, most of the paddy soils belong to the Cambisols or Gleysols.

The Chinese Soil Taxonomy (CST) (CRGCST, 2001) defines the order of Anthrosols with the two suborders of Stagnic Anthrosols (paddy soils) and Orthic Anthrosols (all other Anthrosols). The Stagnic Anthrosols require an anthrostagnic epipedon overlying a hydragric horizon and an anthrostagnic soil moisture regime. Generally, the definition of the anthrostagnic epipedon is similar to that of the anthraquic horizon of WRB, and the hydragric horizons of CRGCST and WRB roughly correspond to each other.

The original soils before paddy cultivation belonged to a broad variety of soil groups worldwide. In the rice-growing areas of Asia, lowland soils like Fluvisols and Gleysols cover relatively small areas, but a large proportion of them are used for paddy cultivation.

In the Zhejiang Province (P.R. China), new farmland was created during the past 2000 years through the reclamation of marine sediments by construction of protective dikes. Immediately after soluble salts were washed out, the land was put into agriculture. Lower areas were used for a cropping sequence of flooded paddy rice in summer followed by a non-flooded winter crop. The upper areas have been exclusively used for non-flooded crop production, which is predominantly wheat (*Triticum* spp.). None of the soils—paddy and non-paddy—underwent any terrestrial soil formation before agriculture started. This enables us to study the genesis of paddy and non-paddy soils in a unique way, as the on-going land reclamation over the last 2000 years has established a chronosequence of soils of different ages. After desalinization the point zero of soil formation is identical for both chronosequences.

The processes of paddy soil formation are triggered by the paddy management. A first clue to soil properties in five soils of this chronosequence was given by Chen et al. (2011), who worked with 50-, 300-, 700-, and 1000-year-old paddy soils. The authors attempted to identify pedogenic thresholds for the different soil processes of the paddy chronosequence up to 1000 years. They were able to identify a major threshold that occurred between 300 and 700 years that was mainly associated with removal of carbonates from the soil profile. Some of the postulated properties change rapidly within 50 years (accumulation of soil organic carbon [SOC] in the surface horizon and decreasing concentrations of Ca, Mg, and Na), other properties change relatively slowly within 700 years (decalcification and migration of clay, P, Mn, and Fe) and some properties change virtually not at all (clay mineral composition and Si, Al, and K concentrations). Yet, the overall rates of soil formation processes remained unclear, and no comparison was made in regard to soil formation occurring under dryland management conditions.

Our objectives were to find answers to the questions: (i) How rapidly or at which timeframes does paddy soil formation occur on these marshland soils? (ii) Is this specific paddy soil development in accordance with the present classification approach of such soils according to WRB? (iii) How strongly does paddy management change soil development in comparison to soils managed under dryland cropping?

The separation of the reclaimed land into paddy fields and non-flooded fields enabled a direct comparison between the two types of soil genesis. This allowed us to evaluate which pedogenic, minerogenic, microbial, and biogeochemical characteristics of the soils were specific to paddy management and could be used to distinguish different stages of pedogenesis. For this purpose, each soil type of the chronosequence was characterized by a complete soil description according to FAO (2006). We complemented basic data of the soil inventory, i.e., BD, pH and electrical conductivity (EC) values, inorganic element composition, SOC, and more specific data like organic lipid biomarker (total lipid extract, *n*-alkane distribution), as well as microbial C and N (C_{mic}, N_{mic}) and DNA extraction to reconstruct and confirm the sedimentation and land use history of the area.

2. Sites and methods

2.1. Study sites, landscape history, and parent material

The study sites are located near Cixi (30°10′N, 121°14′E), approximately 180 km south of Shanghai and 150 km east of Hangzhou, China. Altitudes in the area within the flood plain range from 2.6 m to 5.7 m above sea level (Zhang et al., 2004). The climate is classified as subtropical with periodical monsoon rain. The mean annual temperature is 16.3 °C and the mean annual precipitation is 1325 mm, reaching highest values in April to October (Cheng et al., 2009). The total evaporation is 1000 mm. Therefore, inundation is needed to maintain standing water during rice growth. More detailed information regarding ground water table, geography, and geochemistry of the study area is given by Cheng et al. (2009).

In the Cixi coastal region of the Zhejiang Province, where land reclamation has been in effect for at least 2000 years, the substrate for all paddy and non-inundated non-paddy soils is predominantly marine tidal flat sediment. These coastal marine sediments are derived from the Yangtze River (Changjiang) and its 1.8 Mkm^2 drainage area, which are the major sources of terrestrial material to the East China Sea. The Yangtze Delta front has progressed more than 250 km since 6000 years BP, whereby about 2000 years BP the progradation rate increased abruptly from 38 to 80 km per 1000 years (Kasuaki et al., 2001) caused

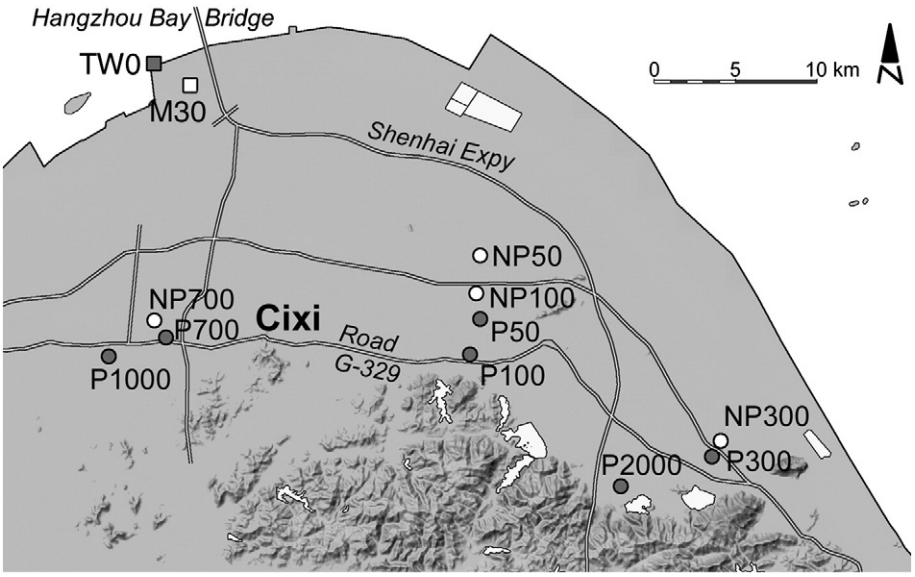


Fig. 1. Sampling locations of the chronosequence area (Kalbitz et al., 2013, modified): tidal wetland (TW0); 30 years old marsh land (M30); paddy soils: P50–P2000; non-paddy soils: NP50–NP700. Main roads are indicated. Road G-329 is almost identical with the main dyke built 1000 years ago.

by high sediment production in the upper reaches due to human activities, such as deforestation and intensive cultivation, and therefore a high sediment load of the Yangtze. Coast-parallel waters flow from the Yangtze estuary directly southwards along the Zhejiang coast where they redistribute hinterland sediments. The high sediment supply to the coastal areas north and east of Cixi, in combination with a drop in sea level, allowed for the intensive land reclamation starting at that time. The marine substrates on which paddy and non-paddy soil in the study area developed are thus preferentially re-deposited terrestrial soils and surface sediments. Thus, the organic and mineral matter composition of the parent material in the study area is predominantly derived from soil and sediment erosion in the Yangtze hinterland.

2.2. Land management for paddy rice cultivation

Paddy fields are constructed so that water may flow under gravity from higher to lower fields; the bunds and outlets regulate the water depth so it does not exceed a few centimeters (Greenland, 1997). In the Cixi area paddy fields have a mean size of 2000 to 3000 m^2. In Asian countries puddling is the most common practice of land preparation for rice cultivation. It breaks down stable aggregates (Ahmad, 1993) and results in a minimum percentage of air-filled pores (Koenigs, 1963). Repeated puddling done by water buffalo or machines over a long period leads to the desired compacted plow pan, reducing permeability and increasing the plant available water capacity (Janssen and Lennartz, 2006). The puddling practice is followed by transplanting rice plants and growing the crop in a submerged environment from crop establishment until close to harvest (Gong, 1983; Sahrawat, 2005; http://www.knowledgebank.irri.org/rkb/watermanagement.html). The fields are usually drained 10–15 days before the expected harvest date. This accelerates maturity and grain ripening, prevents excessive nitrogen uptake, and makes the land easily accessible for harvest operations (http://www.knowledgebank.irri.org/rkb/watermanagement.html; Gong, 1983; Sahrawat, 2005; Kögel-Knabner et al., 2010). Landscape and seasonal changes in rainfall interact in determining the length of time that paddy soils are flooded and the depth to which they are flooded (Greenland, 1997).

In the Yangtze River Valley as well as in the northern part of the Zhejiang Province, much of the land is planted to a rice–wheat or other non-inundated winter upland crop rotation (Cao and Zhang, 2004; Lin et al., 2004). The investigated soils in the Cixi region are used for paddy rice cultivation in the flooding season from April to October, and the local rice cropping system is single-crop rice (Zou et al., 2011). Common rice intercrops planted in the winter season are wheat, oilseed rape, barley, broad bean, and vegetables (Cheng et al., 2009). Other parts of the land are used exclusively for non-flooded crop production.

Paddy soils receive organic manuring (animal manure, rice straw and other crop residues, often fermented with sediments taken from the river or channel, but also charred residues), liming, and fertilization (Cao et al., 2006; Gong, 1983; Kögel-Knabner et al., 2010). In the main rice-growing regions of China, the amount of organic manure applied has doubled since 1952, and it is assumed that organic manure and chemical fertilizers complement each other in intensive agriculture (Wen, 1984).

2.3. Sampling

In a sampling campaign in June 2008, soil profiles were excavated and the soils were completely described according to FAO (2006). The duration of paddy or non-paddy management of the soils (Fig. 1) was estimated according to the County Annals of the Zhejiang Province, considering the time of embankment and dike construction (see also Cheng et al., 2009, for more details). Information in Chinese is obtainable at http://www.cixi.gov.cn/ (Cheng et al., 2009). Soils older than 700 years were all used for paddy rice cultivation. One profile was situated at the mudflat (estuarine sediment; tidal wetland [TW0]), representing the parent material (day 0 of terrestrial soil development); another one was nearby in a 30-year-old marsh land (M30) which has not been under agricultural use so far. All other soil profiles represented up to six ages of the chronosequence (50, 100, 300, 700, 1000 and 2000 years of paddy use [P50–P2000] and 50, 100, 300 and 700 years of land use without lowland rice [non-paddy, NP50–NP700]; Fig. 1). P50 was situated on a site that had been reclaimed with one of the older dikes and was originally used for non-paddy crop cultivation. Fifty years ago, several meters of non-paddy soil were excavated and paddy use started.

At every site, three profiles on different fields were established, with a distance of at least 50 m from each other and sampled in comparable depth increments guided by the horizon boundaries.

Undisturbed soil cores (100 cm^3, in triplicate) of each horizon were taken under field-moist conditions to determine the BD according to Blake and Hartge (1986). In addition, composite samples were taken

for each horizon from different parts of the profile pit and air-dried; subsamples for biomass analyses were stored at 4 °C. All soil samples were sieved to a size of <2.0 mm for further analyses.

2.4. Field analysis of redox conditions

Redox (Pt-)electrodes were installed horizontally into the soil profile walls. Electrodes were installed 10 and 35 cm below the surface at the 100-, 700-, and 2000-year-old paddy and non-paddy sites (P100 and NP100; P700 and NP700; P2000); all sites were equipped with five electrodes per depth. After installation, profiles were carefully filled to approximately restore original conditions. Eh dynamics were monitored during one entire paddy-winter crop cycle (March 2009 to March 2010). Redox measurements were done manually (daily after flooding and biweekly after drainage of the fields), using a portable pH/Eh-meter. Redox data were corrected for the potential of a standard hydrogen electrode by adding 213 mV (20 °C) to the experimental values.

2.5. Laboratory analyses

Particle-size distribution of the <2 mm fraction was determined after dissolution of carbonates with HCl, removal of organic matter with H_2O_2, and dispersion with $Na_4P_2O_7$. The sand content was obtained via wet sieving with a >63 μm screen, and silt and clay were determined by sedimentation rate and X-ray absorption (Sedigraph 5100, Micromeritics GmbH, Aachen, Germany).

Soil pH was measured with a combined electrode in water and 1 M potassium chloride at a soil/solution ratio of 1:2.5 (Schlichting et al., 1995). The suspension was shaken for 5 min and allowed to equilibrate for 1 h. The same procedure was used to measure the electrical conductivity ($EC_{1:2.5}$) and transformed to the "saturation extract" (EC_{SE}) according to FAO (2006).

Total carbon concentration of each soil horizon was determined by dry combustion at 950 °C on a Vario EL elemental analyzer (Elementar Analysensysteme, Hanau, Germany). The inorganic carbon (IC) content was determined by dissolution of carbonates with 42% phosphoric acid and subsequent infrared detection of the evolving CO_2 (C-MAT 550, Ströhlein GmbH, Viersen, Germany). Organic carbon (OC) was calculated by subtracting the concentration of IC from total carbon. OC as well as IC data are already published in Wissing et al. (2011) and Kalbitz et al. (2013). A detailed discussion about different measuring procedures applied to selected samples of the chronosequence as well as their respective measuring inaccuracies can be found in Kalbitz et al. (2013).

Cation exchange capacity (CEC) was determined by saturating 5 g of soil with 1 M sodium acetate at pH 7 and replacing sodium by subsequent washing with 1 M ammonium acetate at pH 7 according to Bower et al. (1952). Oxalate-extractable Al, Fe, Mn, and Si (abbreviated as Al_o, Fe_o, Mn_o, and Si_o) were determined according to Blakemore et al. (1987) with NH_4-oxalate at pH 3.0 after shaking for 2 h in the dark.

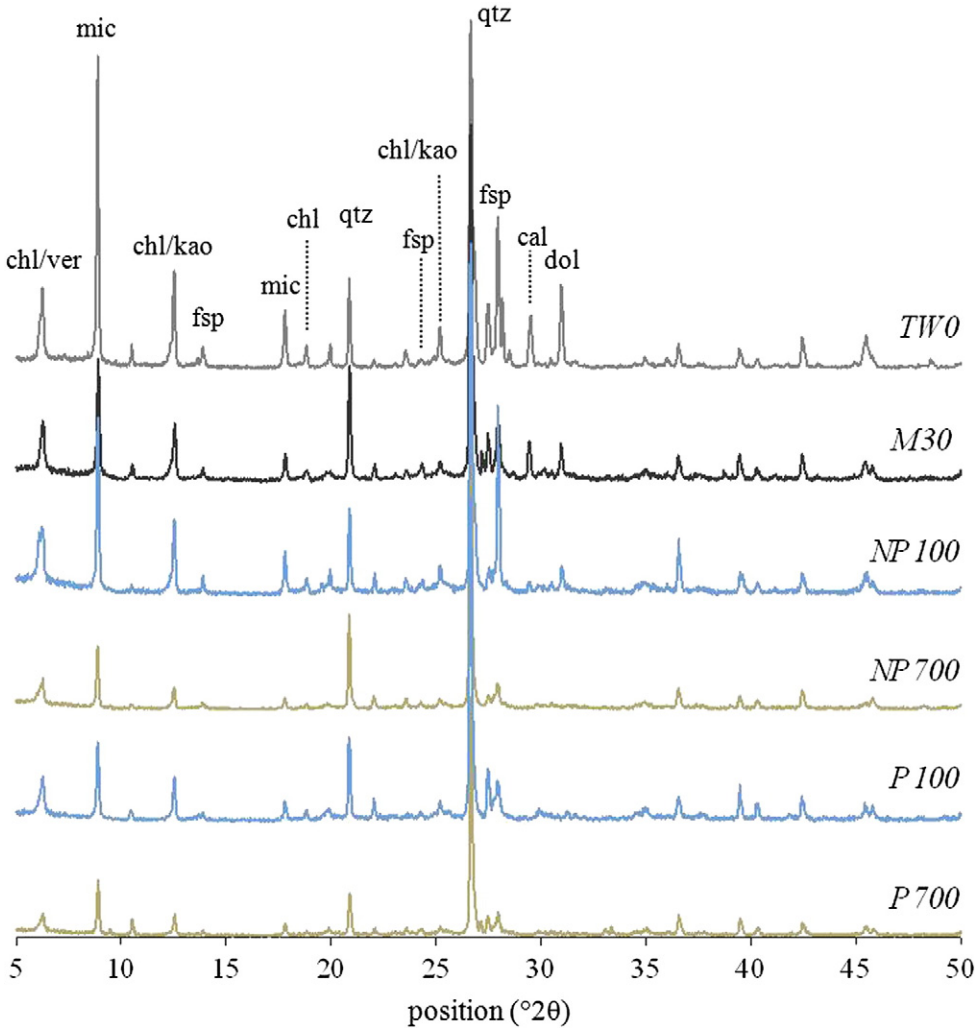


Fig. 2. XRD powder pattern (CuKa) of <2 mm fraction of the reference sites, tidal flat (TW0) and marshland (M30), the Ap-horizons of the non-paddy sites NP100 and P700 and the Alp-horizons (puddled layer) of the paddy sites P100 and P700. Cal: calcite; chl: chlorite; dol: dolomite; fsp: feldspars; kao: kaolinite; mic: mica; qtz: quartz; ver: vermiculite.

Total "free" oxides of Al, Fe, Mn and Si were determined by dithionite–citrate–bicarbonate (DCB) extraction after Mehra and Jackson (1960) (abbreviated as Al_d, Fe_d, Mn_d, and Si_d). The concentrations of Al_o, Fe_o, Mn_o, and Si_o as well as Al_d, Fe_d, Mn_d, and Si_d were determined by inductively coupled plasma-optical emission spectroscopy. All analyses were done in duplicate or triplicate. Total content of elements was determined by X-ray fluorescence (SRS 3000, Siemens AG, Karlsruhe, Germany). The mineral assemblage of the fine earth (<2 mm) was determined on powder specimens using X-ray diffraction (D5005, Siemens AG, Karlsruhe, Germany, with Bragg-Brentano-Geometry) with Cu Kα-radiation (40 kV, 30 mA).

Thin sections of undisturbed samples of the subsoils of P700, P1000, and P2000 were prepared according to Ashley (1973) and micromorphologically studied according to Bullock et al. (1985).

For the determination of *n*-alkanes, soil samples (8 g) were extracted by pressurized solvent extraction (DIONEX ASE-200) at 75 °C and 50 bar, using a binary mixture of dichloromethane/methanol (93:7, v/v). The total extract was fractionated via solid phase extraction (500 mg, activated silica 100 mesh) using hexane (10 ml) to elute aliphatic hydrocarbons. Aliphatic hydrocarbons were analyzed by GC/MS (HP5972 coupled to 5890-II) in EI mode (70 eV) scanning masses *m/z* 50 to 550. Samples were injected in splitless mode onto a 30 m HP5-MS column (30 m × 0.25 mm × 0.25 μm) using an oven program of 70 °C isothermal for 3 min, followed by a ramp of 10 °C/min to 140 °C, followed by 3 °C/min to 330 °C and an isothermal period of 15 min.

Microbial biomass in the top 20 cm of paddy soils under flooded conditions was measured by chloroform fumigation extraction. In brief, samples were fumigated for 24 h in chloroform prior to 0.01 M $CaCl_2$ extraction and subsequent C and N analysis using a DIMATOC 100 total organic carbon analyzer (DIMATEC Analysentechnik GmbH, Essen, Germany). Microbial biomass C and N (C_{mic}, N_{mic}) were calculated as the differences between fumigated and non-fumigated samples (Jörgensen and Brookes, 1991).

Total DNA was extracted from 0.5 g of soil with the Fast DNA Spin kit for soil (MP Biomedicals, Germany) according to the protocol of the manufacturer. Quality and quantity of the DNA extracts were checked with a spectrophotometer (Nanodrop; PeqLab, Germany) according to Schauss et al. (2009). For determination of both microbial biomass and total DNA, five field replicates per site were analyzed.

2.6. Statistical analyses

To evaluate significant differences of measured parameters between soil profiles of different ages on a horizon basis, we used data of similar depth increments. To compare paddy soil profiles, two topsoil and two subsoil horizons were chosen:

1. puddled horizon (Alp) (average depth: 0–16 cm)
2. plow pan (Ardp) (average depth: 16–22 cm)
3. upper B horizon (B1) (average depth: 22–37 cm)
4. lower B horizon (B2) (average depth: 37–50 cm).

To compare entire puddled layers, we recombined subdivided Alp horizons (P50: Alp + Arp; P100, P700 and P1000: Alp1 + Alp2). Ardp horizons of every soil profile were used to compare plow pans. For uppermost B horizons (B1), we used Bwg1 of P50 and P100, Bwdl of P300, Bg of P700, 2Bg of P1000, and Bdg of P2000. For B2, we used Bwg2 of P50 and P100, Bwl of P300, Bg of P700, 2Bg of P1000, and 2Bg1 of P2000.

Since non-paddy soil profiles did not show a distinct plow pan, we compared entire Ap horizons as well as an upper and lower subsoil horizon:

1. plow horizon (Ap) (average depth: 0–17 cm)
2. upper B horizon (B1) (average depth: 17–25 cm)
3. lower B horizon (B2) (average depth: 28–45 cm).

We recombined subdivided Ap horizons (NP50: Ap + ABw; NP100 and NP700: Ap1 + Ap2), and chose B horizons at comparable soil depths.

Mudflat and marshland soils were also included. Assuming a homogeneous mudflat, we used values of TW0 (0–30 cm zone) for comparison at each of the four depth increments. Assuming that soil development had commenced only in the uppermost parts of the M30, we used the 0- to 13-cm zone for comparison with Alp (paddy) and Ap (non-paddy), and the (mudflat-like) 13- to 30-cm zone for comparison with Ardp, B1, and B2.

Due to missing independent field replicates, TW0 and M30 could not be included in MANOVA; however, comparisons between the paddy and non-paddy chronosequence showed differences for most parameters. Therefore, we included these samples in a two-step cluster analysis in order to combine all soils or sediments in significantly different age groups.

All statistical analyses were carried out using SPSS 19.0 (SPSS Inc., Chicago, IL, USA). Interrelationships between soil age and selected parameters (OC, IC, BD, pH, $EC_{1:2.5}$, Fe_d, Fe_o) were analyzed for the 4 (paddy) and 3 (non-paddy) soil depths mentioned above by calculating Spearman's rank correlation coefficients (assuming monotonic, but not necessarily linear relationships between parameters). Mean values of these parameters were horizontally compared with MANOVA, using a Scheffé procedure (for equal variances) and a Dunnett T3 procedure (for unequal variances) as post-hoc tests. To classify the degree of soil development, a two-step cluster analysis procedure was applied using standardized values of the parameters. Finally, structure coefficients derived from a discriminant analysis were used to look for linear combinations of variables that best explained the chronosequence data.

3. Results

3.1. Description of soil profiles

In the tidal wetland and the marshland, the stratification from the marine sedimentation was clearly visible. In the other soils, this stratification was lost due to the formation of soil structure (Fig. S1). With the exception of the tidal wetland, the A horizon was darker than the underlying parts of the soil. In the paddy soils, these A horizons were subdivided into a loose puddled layer overlying a denser plow pan and both were characterized by reduced matrix colors and intensely red oxide concentrations in the root channels. This color pattern, although less pronounced, was also found in the deeper subsoils of all paddy and the younger non-paddy profiles (NP50, NP100). In the older paddy soils (P700, P1000, and P2000), the horizons below the plow pan showed oxide concentrations in the interiors of the aggregates and reduced colors at the aggregate surfaces, with some totally bleached in their upper parts. Buried A horizons appeared in the older paddy soils (P700, P1000, P2000), some of them being very dark.

The soils of the chronosequence belong to the Reference Soil Groups (RSGs) Fluvisols, Cambisols, and Anthrosols (Fig. S1) (IUSS Working Group WRB, 2007). Fluvisols are young soils on alluvial sediments for which the stratification of the original sedimentation is still visible in at least 25% of the soil volume in at least 25 to 50 cm depth. The replacement of the original stratification by a soil structure and the start of carbonate removal rapidly lead to the development of a subsurface cambic horizon in both paddy and non-paddy soils, and the soils must be classified as Cambisols. The non-paddy soils did not reach a more advanced stage of soil formation within 700 years.

All paddy soils fulfilled the requirements of the anthraquic horizon. The hydragric horizon was not yet present in the younger paddy soils (P50, P100, P300) that were classified as Anthraquic Cambisols. However, the hydragric horizon was fully developed in the older paddies (P700, P1000, P2000), which were therefore classified as Hydragric Anthrosols. Chen et al. (2011), who also worked with P50, P300, P700, and P1000, classified all of them as Hydragric Anthrosols according to

Table 1a
Basic soil parameters of tidal wetland (TW0), 30 years old marshland (M30), and paddy soils (P50–P2000).

Horizon	Depth cm	Data from main site						Data from main site and two subsites (mean values ± standard error)						
		Soil color (moist)	Sand %	Silt %	Clay %	Texture class	EC_{SE}[a] mS cm^{-1}	Bulk density[c] g cm^{-3}	OC[b] mg g^{-1}	IC[b] mg g^{-1}	N_t mg g^{-1}	OC/N_t	$EC_{1:2.5}$ µS cm^{-1}	pH KCl
Mudflat TW0														
	0–30	n.d.	8	82	11	Si	47.5	1.31 ± –	5.1 ± –	5.0 ± –	0.32 ± –	16	4180 ± –	8.2 ± –
Marsh M30														
0–13	0–13	n.d.	4	83	14	SiL	1.4	1.31 ± –	10.9 ± –	4.2 ± –	0.76 ± –	14	214 ± –	7.8 ± –
13–30	13–30	n.d.	5	82	12	SiL	2.5	1.31 ± –	5.3 ± –	4.9 ± –	0.29 ± –	18	319 ± –	8.0 ± –
Paddy 50														
Alp	0–7	10YR3/3	4	67	29	SiCL	3.1	1.10 ± 0.06	17.8 ± 0.5	1.4 ± 0.3	1.73 ± 0.09	10	637 ± 100	7.4 ± <0.1
Arp	7–14	10YR3/3	5	65	30	SiCL	3.3	1.22 ± 0.05	14.7 ± 0.3	1.6 ± 0.4	1.40 ± 0.03	11	709 ± 15	7.5 ± <0.1
Ardp	14–23	2.5Y4/2	4	64	32	SiCL	1.8	1.56 ± 0.02	5.7 ± 0.1	2.1 ± 0.8	0.47 ± 0.04	12	379 ± 40	7.5 ± <0.1
Bwg1	23–38	10YR4/3	2	69	28	SiCL	1.7	1.47 ± 0.01	5.1 ± 0.2	2.4 ± 0.6	0.43 ± 0.07	12	343 ± 32	7.5 ± <0.1
Bwg2	38–50	10YR4/3	3	75	22	SiL	2.4	1.47 ± 0.02	5.2 ± 0.2	2.4 ± 0.5	0.41 ± 0.09	13	326 ± 28	7.5 ± <0.1
Bwg3	50–70	10YR4/3	3	67	30	SiCL	1.5	1.45 ± 0.02	5.1 ± 0.1	3.4 ± 0.1	0.40 ± 0.05	13	320 ± 26	7.6 ± <0.1
Blg	70–100	n.d.	3	66	31	SiCL	1.5	1.41 ± 0.01	5.4 ± 0.2	3.8 ± 0.1	0.41 ± 0.03	13	327 ± 29	7.6 ± <0.1
Paddy 100														
Alp1	0–9	10YR4/2	7	65	28	SiCL	1.8	1.05 ± 0.05	17.6 ± 1.3	<0.1	2.01 ± 0.24	9	422 ± 41	5.0 ± 0.2
Alp2	9–15	10YR4/2	4	68	28	SiCL	1.7	1.25 ± 0.04	15.3 ± 1.1	<0.1	1.82 ± 0.21	8	351 ± 10	5.8 ± 0.1
Ardp	15–21	10YR4/3	5	67	28	SiCL	1.7	1.56 ± 0.02	6.6 ± 0.9	0.2 ± 0.2	0.84 ± 0.20	8	303 ± 9	7.0 ± 0.2
Bw g1	21–30	10YR4/3.5	2	65	33	SiCL	1.8	1.55 ± 0.00	5.8 ± 0.7	1.9 ± 0.6	0.55 ± 0.11	11	323 ± 10	7.4 ± <0.1
Bw g2	30–50	10YR4.5/3	2	65	33	SiCL	1.7	1.53 ± 0.03	4.7 ± 0.2	2.9 ± 0.3	0.43 ± 0.02	11	323 ± 8	7.4 ± 0.2
Bw lg1	50–75	10YR4.5/3	2	68	29	SiCL	1.2	1.49 ± 0.02	4.7 ± 0.1	2.8 ± <0.1	0.40 ± 0.01	12	260 ± 20	7.5 ± 0.1
Bw lg2	75–100	10YR4.5/3	3	69	29	SiCL	1.1	1.39 ± 0.06	5.2 ± 0.1	2.6 ± 0.4	0.42 ± 0.01	12	243 ± 18	7.5 ± <0.1
Paddy 300														
Alp	0–18	10YR4/2	6	70	25	SiL	2.3	1.17 ± 0.05	22.6 ± 2.5	<0.1	2.46 ± 0.38	9	399 ± 100	5.8 ± 0.3
Ardp	18–24	10YR4/2	3	68	29	SiCL	1.4	1.50 ± 0.04	14.9 ± 3.3	<0.1	1.42 ± 0.50	11	304 ± 34	6.6 ± 0.3
Bwdl	24–30	10YR4,5/3	4	72	24	SiL	2.2	1.61 ± 0.04	7.1 ± 1.3	0.6 ± 0.4	0.78 ± 0.34	9	262 ± 12	7.4 ± 0.2
Bwl	30–50	10YR5/3	3	68	28	SiCL	1.4	1.63 ± 0.02	5.2 ± 0.1	0.5 ± 0.3	0.47 ± 0.04	11	270 ± 12	7.4 ± 0.1
Bw lg1	50–70	10YR5/3	6	63	31	SiCL	1.6	1.48 ± 0.03	4.9 ± 0.1	1.1 ± 0.3	0.47 ± 0.02	10	301 ± 3	7.5 ± <0.1
Bw lg2	70–100	10YR5/3	4	63	33	SiCL	1.7	1.46 ± 0.03	4.7 ± 0.2	1.8 ± 1.0	0.46 ± 0.05	10	295 ± 24	7.3 ± 0.2
Paddy 700														
Alp1	0–10	2,5Y 3/2	3	68	29	SiCL	2.2	1.06 ± 0.02	23.2 ± 1.9	<0.1	2.21 ± 0.30	11	533 ± 20	6.7 ± 0.1
Alp2	10–16	5Y 2/2	4	69	28	SiCL	2.5	1.20 ± 0.02	18.0 ± 1.8	<0.1	1.83 ± 0.21	10	480 ± 96	6.6 ± 0.1
Ardp	16–22	2,5 Y 3/2	5	66	28	SiCL	2.5	1.46 ± 0.03	8.6 ± 0.5	<0.1	1.08 ± 0.09	8	468 ± 9	6.7 ± 0.1
Bg	22–45	2,5 Y 4/2	3	69	28	SiCL	1.9	1.49 ± 0.01	4.1 ± 0.6	<0.1	0.55 ± 0.14	7	318 ± 30	6.7 ± <0.1
2Ahgb	45–69	2,5 Y 3/1	2	49	49	SiC	1.1	1.53 ± 0.04	5.3 ± 0.6	<0.1	0.66 ± 0.11	8	272 ± 35	6.5 ± <0.1
2Blg	69–106	5Y 4/1	2	49	49	SiC	1.2	1.45 ± 0.01	3.3 ± 0.4	<0.1	0.52 ± 0.05	6	266 ± 34	6.4 ± <0.1
3Ahlgb	106–116	2,5Y3/1	1	71	28	SiCL	1.3	1.50 ± 0.05	6.0 ± 0.9	<0.1	0.42 ± 0.04	14	251 ± –	6.5 ± –
3Blg	116–130	4Y4/1	1	75	24	SiL	5.8	1.47 ± –	2.1 ± –	<0.1	0.32 ± –	7	649 ± –	6.4 ± –
Paddy 1000														
Alp	0–10	10YR3/3	1	71	28	SiCL	1.0	1.15 ± 0.02	14.0 ± 0.6	<0.1	1.60 ± 0.15	9	181 ± 29	5.2 ± 0.2
Al(d)p1	10–16	10YR3/3	3	70	27	SiL	1.4	1.34 ± 0.02	11.0 ± 0.3	<0.1	1.33 ± 0.06	8	185 ± 13	5.8 ± 0.3
Aldp2	16–21	2,5Y R4/2	3	66	31	SiCL	1.0	1.50 ± 0.04	6.1 ± 0.6	<0.1	0.79 ± 0.11	8	195 ± 11	6.5 ± 0.1
2Ahgb	21–40	2,5Y R3/2	2	57	41	SiC	0.7	1.52 ± 0.02	4.6 ± 0.4	<0.1	0.59 ± 0.07	8	201 ± 9	6.8 ± <0.1
2Bg	40–55	n.d.	3	48	49	SiC	0.8	1.43 ± 0.02	2.8 ± 0.2	<0.1	0.49 ± 0.03	6	233 ± 23	6.7 ± 0.1
2Bl	55–80	n.d.	3	48	49	SiC	0.7	1.29 ± 0.01	3.9 ± 0.1	<0.1	0.61 ± 0.02	6	265 ± 46	6.6 ± 0.1
3Ahlb	80–93	5Y2/1	1	55	44	SiC	0.6	1.41 ± –	15.4 ± –	<0.1	1.39 ± –	11	168 ± –	6.5 ± –
3Bl	93–110	n.d.	4	57	39	SiCL	1.0	1.41 ± –	3.5 ± –	<0.1	0.48 ± –	7	188 ± –	6.5 ± –
Paddy 2000[c]														
Alp	0–15	2,5Y4/1	8	68	24	SiL	2.1	1.02 ± 0.04	30.0 ± 1.5	<0.1	3.85 ± –	8	382 ± –	5.1 ± –
Ar(d)p	15–20	2,5Y4/1	6	69	26	SiL	1.2	1.31 ± 0.14	20.5 ± 3.2	<0.1	3.01 ± –	7	222 ± –	5.5 ± –

Bdg	20–27	2,5Y5/2	4	68	28	SiCL	0.7	1.55 ± 0.02	5.5 ± 0.2	<0.1	0.64 ± –	9	128 ± –	6.5 ± –
2AhgB	27–35	2,5Y4/1	3	61	36	SiCL	0.6	1.57 ± 0.04	3.5 ± 0.2	<0.1	0.44 ± –	8	115 ± –	6.3 ± –
2Bg1	35–50	2,5Y5/1	3	50	47	SiC	0.5	1.49 ± 0.05	2.6 ± 0.4	<0.1	0.38 ± –	7	121 ± –	6.3 ± –
2Bg2	50–70	2,5Y6/2	2	61	38	SiCL	0.5	1.48 ± 0.03	1.6 ± 0.1	<0.1	0.32 ± –	5	95 ± –	6.1 ± –
2Blg	70–100	2,5Y4/2	6	62	32	SiCL	0.4	1.45 ± 0.01	1.4 ± <0.1	<0.1	0.32 ± –	4	77 ± –	6.1 ± –

Horizon	Data from main site and two subsites (mean values ± standard error)									Data from main site							
	DCB extractable		Oxalate extractable		Fe_o/Fe_d	Fe_d/Fe_t	CEC_{pot}	BS	Na saturat.	Total element concentration							
	Fe	Mn	Fe	Mn						Si_t	Ca_t	Mg_t	Na_t	K_t	Al_t	Fe_t	Mn_t
	mg g^{-1}		mg g^{-1}				$mmol_c$ kg^{-1}	% of CEC_{pot}	% of CEC_{pot}	mg g^{-1}							
Mudflat TW0																	
	7.38 ± –	0.28 ± –	4.03 ± –	0.2 ± –	0.55	0.21	79 ± –	>100	>100	329	32.1	14.8	10.2	20	66	35	0.76
Marsh M30																	
0–13	7.99 ± –	0.31 ± –	3.39 ± –	0.3 ± –	0.42	0.26	96 ± –	>100	3.7	331	31.8	16.1	12.2	19	66	31	0.66
13–30	6.64 ± –	0.24 ± –	2.78 ± –	0.2 ± –	0.42	n.d.	71 ± –	>100	13.8	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
Paddy 50																	
Alp	8.50 ± 0.01	0.44 ± 0.01	4.18 ± 0.35	0.3 ± 0.01	0.49	0.23	182 ± 5	>100	8.9	322	17.1	15.8	9.1	22	81	38	0.76
Arp	8.33 ± 0.02	0.45 ± 0.01	3.87 ± <0.01	0.3 ± <0.01	0.46	0.21	172 ± 2	>100	9.9	316	20.5	16.3	9.3	23	83	40	0.87
Ardp	9.92 ± 0.21	0.54 ± 0.02	1.86 ± 0.32	0.4 ± 0.04	0.19	0.24	158 ± 5	>100	7.8	310	23.8	16.7	9.5	23	85	41	0.97
Bw g1	10.29 ± 0.60	0.63 ± 0.09	2.29 ± 0.34	0.6 ± 0.06	0.22	0.26	158 ± 1	>100	7.3	313	24.8	16.5	9.5	23	82	40	1.04
Bw g2	9.21 ± 0.15	0.58 ± 0.04	2.64 ± 0.24	0.5 ± 0.08	0.29	0.24	151 ± 6	>100	7.0	318	25.8	15.7	9.7	22	79	38	0.97
Bw g3	9.16 ± 0.37	0.56 ± 0.02	3.17 ± 0.79	0.5 ± 0.06	0.35	0.24	147 ± 4	>100	7.1	318	25.8	15.6	9.8	22	79	38	0.97
Blg	8.92 ± 0.19	0.47 ± 0.08	3.13 ± 0.07	0.4 ± 0.03	0.35	0.23	149 ± 2	>100	7.4	315	27.1	15.7	9.4	22	80	39	0.72
Paddy 100																	
Alp1	8.43 ± 0.27	0.20 ± 0.01	3.11 ± 0.34	0.1 ± 0.02	0.37	0.25	174 ± 2	72	5.5	341	7.9	11.6	9.7	19	74	34	0.53
Alp2	8.96 ± 0.09	0.23 ± 0.03	3.17 ± 0.11	0.2 ± 0.03	0.35	0.26	179 ± 7	76	5.9	339	8.2	11.7	10.6	19	74	35	0.55
Ardp	10.13 ± 0.48	0.77 ± 0.04	1.53 ± 0.31	0.6 ± 0.10	0.15	0.28	174 ± 1	91	6.3	335	8.7	12.5	10.0	21	77	36	1.24
Bw g1	10.30 ± 0.84	0.78 ± 0.05	1.25 ± 0.13	0.7 ± 0.09	0.12	0.25	165 ± 10	>100	6.6	319	21.0	14.9	9.2	22	79	41	1.14
Bw g2	10.77 ± 0.28	0.63 ± 0.06	1.47 ± 0.22	0.5 ± 0.09	0.14	0.26	160 ± 10	>100	6.1	314	22.4	15.3	8.9	22	83	42	1.04
Bw lg1	10.25 ± 0.16	0.56 ± 0.07	2.08 ± 0.50	0.4 ± 0.09	0.20	0.24	158 ± 6	>100	4.6	318	23.3	15.7	8.9	22	76	43	1.10
Bw lg2	10.90 ± 0.26	0.67 ± 0.03	2.37 ± 0.25	0.6 ± 0.08	0.22	0.27	160 ± 2	>100	4.1	320	23.3	15.0	9.2	22	78	41	1.00
Paddy 300																	
Alp	8.86 ± 0.15	0.19 ± 0.03	2.88 ± 0.19	0.1 ± 0.02	0.33	0.26	195 ± 5	71	5.8	343	8.6	11.5	9.7	19	71	35	0.50
Ardp	9.28 ± 0.48	0.22 ± 0.06	2.22 ± 0.63	0.1 ± 0.01	0.24	0.26	174 ± 6	79	6.8	336	8.4	12.4	10.3	20	77	35	0.52
Bw dl	9.89 ± 0.68	0.44 ± 0.15	0.99 ± 0.11	0.3 ± 0.11	0.10	0.28	138 ± 12	>100	6.1	337	17.1	13.0	10.0	20	68	36	1.11
Bw l	10.03 ± 0.61	0.51 ± 0.06	0.89 ± 0.04	0.3 ± 0.03	0.09	0.27	138 ± 4	>100	6.4	329	12.7	15.2	9.6	22	77	37	0.71
Bw lg1	10.36 ± 0.22	0.54 ± 0.02	1.49 ± 0.19	0.5 ± 0.05	0.14	0.25	167 ± 7	>100	6.7	315	16.5	16.3	8.8	24	85	42	0.88
Bw lg2	10.79 ± 0.55	0.57 ± 0.11	1.85 ± 0.50	0.6 ± 0.15	0.17	0.25	169 ± 5	>100	7.3	313	20.0	16.2	8.7	24	83	43	1.04
Paddy 700																	
Alp1	8.26 ± 0.29	0.25 ± 0.01	4.59 ± 0.14	0.2 ± 0.01	0.56	0.24	215 ± 14	82	6.1	339	10.3	12.5	9.4	20	73	34	0.59
Alp2	7.68 ± 0.59	0.21 ± 0.03	4.07 ± 0.13	0.2 ± 0.02	0.53	0.23	197 ± 11	85	7.1	344	9.1	11.2	9.9	19	71	33	0.55
Ardp	7.79 ± 0.86	0.23 ± 0.04	2.61 ± 0.15	0.2 ± 0.01	0.34	0.24	166 ± 7	84	7.5	344	7.6	10.9	10.8	20	71	33	0.52
Bg	7.71 ± 0.49	0.42 ± 0.12	1.51 ± 0.16	0.4 ± 0.11	0.20	0.25	140 ± 10	84	8.1	346	6.8	10.4	10.5	20	71	31	0.89
2Ahgb	14.70 ± 3.62	0.21 ± 0.04	1.38 ± 0.66	0.2 ± 0.03	0.09	0.34	204 ± 26	77	7.4	315	5.6	12.1	8.1	25	95	43	0.46
2Blg	20.41 ± 4.78	0.29 ± 0.13	1.25 ± 0.26	0.2 ± 0.08	0.06	0.34	231 ± 24	73	7.5	306	5.1	12.8	7.8	25	91	59	0.56
3Ahlgb	3.94 ± –	0.06 ± –	0.85 ± –	0.04 ± –	0.22	0.16	170 ± –	77	9.4	352	7.1	9.7	11.2	19	70	25	0.35
3Blg	7.00 ± –	0.09 ± –	0.93 ± –	0.07 ± –	0.13	0.23	130 ± –	83	12.4	349	6.6	11.2	11.4	20	68	31	0.41
Paddy 1000																	
Alp	8.34 ± 0.28	0.26 ± 0.02	3.71 ± 0.33	0.2 ± 0.01	0.44	0.27	171 ± 8	69	3.7	350	7.1	9.2	10.2	19	69	31	0.51
Al(d)p1	8.24 ± 0.33	0.26 ± 0.04	3.55 ± 0.20	0.2 ± 0.04	0.43	0.27	171 ± 9	74	4.4	351	7.3	9.2	10.2	19	68	30	0.52
Aldp2	8.84 ± 0.67	0.62 ± 0.17	1.70 ± 0.26	0.6 ± 0.08	0.19	0.28	159 ± 2	80	5.3	346	6.4	9.7	10.0	20	73	32	0.94
2Ahgb	8.18 ± 0.61	0.19 ± 0.03	0.56 ± 0.08	0.2 ± 0.06	0.07	0.22	197 ± 14	79	5.2	325	5.7	11.2	8.4	24	89	38	0.48
2Bg	29.21 ± 4.70	0.25 ± 0.05	1.76 ± 0.40	0.2 ± 0.04	0.06	0.41	226 ± 10	78	5.4	304	5.4	12.0	6.8	24	86	71	0.68

(continued on next page)

Table 1a (*continued*)

Horizon	Data from main site and two subsites (mean values ± standard error)									Data from main site							
	DCB extractable		Oxalate extractable		Fe_o/Fe_d	Fe_d/Fe_t	CEC_{pot}	BS	Na saturat.	Total element concentration							
	Fe	Mn	Fe	Mn						Si_t	Ca_t	Mg_t	Na_t	K_t	Al_t	Fe_t	Mn_t
	mg g^{-1}		mg g^{-1}				$mmol_c$ kg^{-1}	% of CEC_{pot}	% of CEC_{pot}	mg g^{-1}							
2Bl	23.16 ± 5.98	0.39 ± 0.12	2.59 ± 0.18	0.4 ± 0.08	0.11	0.42	282 ± 8	74	5.4	309	5.7	12.6	6.9	27	91	55	0.68
3Ahlb	4.56 ± –	0.14 ± –	1.66 ± –	0.10 ± –	n.d.	n.d.	311 ± –	n.d.	n.d.	328	8.0	12.1	8.5	21	91	30	0.46
3Bl	12.63 ± –	0.34 ± –	2.85 ± –	0.32 ± –	n.d.	n.d.	190 ± –	n.d.	n.d.	329	6.6	12.4	8.3	22	80	42	0.71
Paddy 2000[c]																	
Alp	5.95 ± –	0.13 ± –	3.13 ± –	0.1 ± –	0.52	0.27	203 ± –	58	3.2	355	6.6	7.0	9.9	18	73	22	0.37
Ar(d)p	6.59 ± –	0.08 ± –	4.47 ± –	0.1 ± –	0.68	0.28	182 ± –	63	2.3	362	6.5	6.2	9.9	18	65	23	0.35
Bdg	16.89 ± –	0.62 ± –	1.38 ± –	0.3 ± –	0.08	0.51	136 ± –	73	<1	361	5.6	5.3	10.5	17	61	33	0.70
2AhgB	9.48 ± –	0.91 ± –	0.67 ± –	1.4 ± –	0.07	0.33	172 ± –	74	1.9	346	5.7	7.7	9.1	19	78	29	1.32
2Bg1	24.82 ± –	0.32 ± –	0.44 ± –	0.2 ± –	0.02	0.41	202 ± –	72	1.9	322	5.3	9.0	7.5	20	80	60	0.63
2Bg2	15.35 ± –	0.28 ± –	1.00 ± –	0.2 ± –	0.07	0.34	199 ± –	72	1.7	333	6.2	10.4	7.8	21	77	46	0.64
2Blg	14.19 ± –	0.44 ± –	2.47 ± –	0.4 ± –	0.17	0.32	173 ± –	83	1.9	336	7.1	10.5	7.8	20	73	44	0.89

n.d.: not determined.

[a] Calculated from $EC_{1:2.5}$ according to FAO (2006).

[b] Data from Wissing et al. (2011).

[c] Most data only from main site.

WRB. However, the hydragric horizon was absent in P50 and P300, and the other applying qualifiers were not mentioned.

The Tidalic qualifier indicates the tidewater influence and was only applicable to the tidal wetland (TW0). P700 had a buried mollic horizon, which is expressed by the Thaptomollic qualifier. The Eutric qualifier expressed a high base saturation (≥50%). It applied to all our soils, but to avoid redundant qualifiers, it was only used for soils that were neither Calcaric nor (Hypo-)Sodic. The Calcaric, (Hypo-)Sodic, and Eutric qualifiers emphasized the generally low leaching (lixiviation) of base cations and the Eutric qualifier also the high soil fertility. All soils had the Siltic qualifier, which encompasses the textural classes (FAO, 2006) of silt, silt loam, silty clay loam, and silty clay. All soils were influenced by ascending groundwater. In the paddy soils as well as in NP50, NP100, and the marshland, a gleyic color pattern was expressed strongly enough to account for the Gleyic qualifier. If it was only present below 50 cm, it was further specified as Endogleyic.

3.2. Texture and mineralogy

Throughout the chronosequence, topsoil texture was dominated by silt (approx. 65%–75%), followed by clay (approx. 20%–30%). The youngest sediments (M30 and TW0) tended to have a slightly coarser texture, showing silt contents >80%. Buried horizons of the older paddy soils were characterized by a finer texture (up to 49% clay). The micromorphological studies (data not shown) did not detect any clay cutans.

The sediments contained quartz, feldspars (i.e., plagioclase, little orthoclase), mica, and calcite as well as dolomite. Clay minerals present were kaolinite, chlorite and vermiculite (Fig. 2). Only modest differences in peak height for quartz, feldspars, mica, and the clay minerals kaolinite, chlorite and vermiculite could be observed upon cultivation (Fig. 2), without any apparent trend for gains or losses either with age of the soils or between paddy and non-paddy soils.

3.3. Bulk density

The BDs (Tables 1a, 1b) of the non-paddy soils ranged from 1.25 to 1.50 kg dm^{-3} with a marked increase from topsoils to subsoils, but without a denser plow pan. All paddy soils had a clearly developed plow pan, with BDs from 1.46 to 1.61 kg dm^{-3}. In the paddy soils ≥100 years, the horizons directly below the plow pan still had high BD values, which were very similar to the plow pan values. The puddled layers above the plow pans showed values between 1.02 and 1.15 kg dm^{-3} in their upper parts and from 1.20 to 1.25 kg dm^{-3} in their lower parts, but without chronological trend. Below the plow pans, BDs remained ≥1.40 kg dm^{-3} with one exception: the P1000 lower BDs from 55 to 80 cm in a layer above a buried A horizon.

3.4. Organic carbon and total nitrogen

Parent materials contained about 5 g kg^{-1} organic carbon (OC). The OC concentrations in the uppermost horizons of the non-paddy topsoils did not depend on the time of soil development and varied between 10.5 and 11.0 g kg^{-1} (Tables 1a, 1b). They gradually decreased to values between 4.2 and 5.3 g kg^{-1} at 1-m depth. In the paddy soils, OC concentrations of the uppermost horizons increased with time of paddy use from 17.8 (P50) to 30.0 g kg^{-1} (P2000) with the striking exception of P1000 (14.0 g kg^{-1}). Organic C concentrations sharply decreased in the plow pan in all paddy soils, followed by a continuous decrease with soil depth. Subsoil OC concentrations decreased with time of soil development from about 5 g kg^{-1} in P50 to less than 2 g kg^{-1} in P2000. Some of the old paddy soils had morphologically recognizable buried A horizons that showed higher OC values (P700, P1000, Table 1a).

Both paddy and non-paddy soils contained significantly more total nitrogen (N_t) than the parent materials. Due to the increase in BD, both soils also accumulated N_t in the surface soils during the first 100 years of land embankment (see also Roth et al., 2011). The N_t concentrations, however, did not depend on time of soil development in the non-paddy fields; they ranged between 1.1 and 1.3 g kg^{-1} and decreased in the subsoils down to 0.4 g kg^{-1} (Table 1b). As for OC, the N_t concentrations in paddy soils showed a more pronounced differentiation between topsoils and subsoils. Total N concentrations in the uppermost horizon increased with time of paddy soil development from 1.7 to 3.9 g kg^{-1}, again with the exception of P1000. Subsoil N_t values were of the magnitude in the non-paddy soils. Except for the 3Ahlb horizon in the P1000 profile, buried A horizons did not clearly show increased N_t concentrations compared to the horizon above and below (Table 1a). OC to N ratios were relatively low, ranging between 4.7 (P2000 subsoil) and 14 (P700 buried A horizon).

3.5. Microbial biomass carbon and nitrogen

Results of C_{mic} and N_{mic} measurements in paddy topsoils at the time point before rice was planted were closely linked to total organic carbon and N_t contents (Roth et al., 2011). Linear regression analysis with logarithmized soil age showed that C_{mic} increased highly significantly ($p < 0.01$) with cultivation time (from 0.72 in P50 to 5.1 g kg^{-1} in P2000) and paralleled amounts of extracted DNA. For N_{mic}, a significant difference was only measured between TW0 and P2000 (Table 3). Consequently, the ratio of C_{mic} to N_{mic} increased with cultivation time of paddy soils. During the vegetation period, however, C_{mic} and N_{mic} were in all soils more influenced by the plant development stage as well as fertilization or plant protection methods than by the time of paddy soil evolution (data not shown).

3.6. Lipid biomarkers

The use of modern reference sediments allowed for characterizing local substrate types of freshwater and marine tidal flat sediments. The limnic substrate exhibited higher amounts of the nC_{23} and the nC_{25} alkanes and equal proportions of nC_{29} and nC_{33} alkanes (Fig. 3). The marine reference sediment exhibited a preponderance of nC_{25} over nC_{23} and of nC_{29} over nC_{33} (Fig. 3). Most subsoils mimicked the *n*-alkane distribution of the marine substrate, and only a few locations showed subsoil horizons that indicated a freshwater substrate, i.e., sedimentation in a coastal lagoon setting (see Fig. 3: P50, 23–38 cm).

Continuous utilization as non-inundated soils or paddy soils introduced crop plant waxes of different composition into the soils. After only 50 years of cultivation the topsoils under non-paddy use had significantly higher proportions of nC_{31} alkanes when compared to preferential paddy soil use (Fig. 4). Increasing chain length of *n*-alkanes resulted in lower water permeability of plant surface wax. Lower evaporative water loss from partially submerged growing rice plants in comparison to upland cereal crops (wheat, maize) and rape is consistent with lower relative nC_{31}-content in paddy soils. The topsoil *n*-alkane pattern thus served as a tool to discriminate between paddy or non-paddy management practices (Fig. 4) and confirmed historical records of land use.

3.7. Carbonates

The sediments of the Cixi region contained up to 5 g kg^{-1} of inorganic carbon (IC) (Table 1a), which corresponds to 4% $CaCO_3$-equivalents. IC concentrations decreased during soil development in both the non-paddy and paddy soils. In NP50, NP100 and NP300, IC was present throughout the profiles; in NP700 it was only in the subsoils below 17 cm. The highest subsoil IC concentrations were found in NP50 with 4.2 g kg^{-1}, which was close to the IC concentration of the sediment in

Table 1b
Basic soil parameters of non-paddy soils (NP50–NP700).

Horizon	Depth cm	Data from main site						Data from main site and two subsites (mean values ± standard error)						
		Soil color (moist)	Sand %	Silt %	Clay %	Texture class	EC_{SE}[a] mS cm^{-1}	Bulk density g cm^{-3}	OC[b] mg g^{-1}	IC[b] mg g^{-1}	Nt mg g^{-1}	OC/N_t	$EC_{1:2.5}$ µS cm^{-1}	pH KCl
Non-paddy 50														
Ap	0–9	10YR4,5/3	4	76	21	SiL	1.8	1.31 ± 0.01	10.6 ± 0.2	1.6 ± 0.2	1.13 ± 0.07	9	317 ± 42	7.3 ± 0.1
ABw	9–17	10YR5/3	3	74	23	SiL	2.7	1.38 ± 0.02	8.2 ± 0.7	2.2 ± 0.2	0.78 ± 0.15	10	377 ± 68	7.3 ± 0.1
Bw	17–24	10YR5/3	2	73	24	SiL	3.6	1.37 ± 0.03	6.4 ± 0.3	3.1 ± 0.2	0.55 ± 0.05	12	417 ± 64	7.5 ± <0.1
BCg	24–45	10YR5/3	8	69	23	SiL	3.8	1.39 ± 0.01	5.6 ± 0.2	3.5 ± 0.1	0.48 ± 0.06	12	459 ± 70	7.5 ± <0.1
CBg	45–70	10YR5/3	3	70	27	SiL	5.7	1.43 ± 0.03	4.8 ± 0.2	4.2 ± 0.3	0.40 ± 0.04	12	596 ± 122	7.6 ± <0.1
CBlg	70–100	10YR5/3	2	75	22	SiL	7.0	1.42 ± 0.01	4.7 ± 0.1	4.2 ± 0.2	0.38 ± 0.03	12	652 ± 153	7.6 ± <0.1
Non-paddy 100														
Ap1	0–14	10YR3/3	4	69	27	SiCL	1.6	1.31 ± 0.02	10.8 ± 1.0	0.7 ± 0.4	1.12 ± 0.22	10	190 ± 57	7.3 ± 0.1
Ap2	14–25	10YR4/3	1	73	26	SiL	2.1	1.37 ± 0.01	8.1 ± 0.3	1.2 ± 0.2	0.79 ± 0.06	10	178 ± 47	7.4 ± <0.1
Bw	25–30	10YR4/3	4	67	29	SiCL	1.5	1.44 ± 0.02	7.0 ± 0.2	2.1 ± 0.4	0.65 ± 0.01	11	191 ± 48	7.4 ± 0.1
BCw g1	30–38	10YR4/4	3	65	32	SiCL	1.9	1.49 ± 0.03	6.2 ± 0.3	2.1 ± 0.2	0.55 ± 0.02	11	211 ± 77	7.5 ± <0.1
BCw g2	38–70	10YR4/4	2	65	33	SiCL	2.1	1.50 ± 0.02	5.5 ± 0.3	2.8 ± 0.5	0.47 ± 0.05	12	255 ± 77	7.5 ± <0.1
BCw lg	70–100	10YR4/4	2	67	31	SiCL	2.2	1.50 ± 0.02	5.3 ± 0.3	3.4 ± 0.5	0.48 ± 0.04	11	312 ± 58	7.5 ± <0.1
Non-paddy 300														
Ap	0–11	10YR4/3	3	74	24	SiL	0.7	1.32 ± 0.01	10.5 ± 0.1	0.1 ± 0.1	1.19 ± 0.05	9	150 ± 24	7.0 ± 0.2
Bw1	11–22	10YR4,5/3	3	71	26	SiL	0.8	1.39 ± 0.01	6.6 ± 0.6	0.2 ± 0.1	0.70 ± 0.09	10	149 ± 27	7.1 ± 0.1
Bw2	22–32	10YR5/3	4	68	29	SiCL	0.5	1.37 ± 0.01	6.2 ± 0.3	0.5 ± 0.1	0.63 ± 0.04	10	157' ± 32	7.2 ± 0.1
Bwg1	32–50	10YR5/3	2	68	31	SiCL	0.7	1.42 ± 0.02	5.3 ± 0.1	2.7 ± 0.3	0.44 ± 0.02	12	182 ± 26	7.5 ± 0.1
Bwg2	50–70	10YR5/3	3	69	28	SiCL	1.5	1.43 ± 0.01	4.6 ± 0.1	4.0 ± 0.1	0.40 ± 0.02	12	249 ± 25	7.5 ± <0.1
Bw lg	70–100	10YR4/3	6	67	28	SiCL	0.8	1.46 ± 0.02	4.4 ± 0.2	3.9 ± 0.2	0.40 ± 0.02	11	234 ± 54	7.6 ± <0.1
Non-paddy 700														
Ap1	0–12	10YR3,5/3	5	76	19	SiL	0.9	1.25 ± 0.06	11.0 ± 1.9	<0.1	1.31 ± 0.40	8	148 ± 9	5.9 ± 0.3
Ap2	12–17	7,5Y R4/3	2	76	22	SiL	1.7	1.32 ± 0.03	7.1 ± 1.8	<0.1	0.87 ± 0.06	8	122 ± 12	6.6 ± 0.1
Bw 1	17–23	7,5Y R4/3	2	73	25	SiL	0.8	1.38 ± 0.01	5.3 ± 0.8	0.3 ± 0.2	0.59 ± 0.17	9	116 ± 12	6.9 ± 0.2
Bw2	23–45	7,5Y R4/3	5	76	19	SiL	1.2	1.39 ± 0.03	5.3 ± 0.4	1.3 ± 0.4	0.44 ± 0.02	12	150 ± 5	7.3 ± 0.1
Bw l1	45–70	7,5Y R4/3	4	81	15	SiL	1.6	1.42 ± 0.02	4.8 ± 0.1	1.9 ± 0.7	0.41 ± 0.03	12	168 ± 11	7.4 ± <0.1
Bw l2	70–100	7,5Y R4/3	4	73	23	SiL	2.0	1.42 ± 0.01	4.2 ± 0.5	2.2 ± 0.5	0.41 ± 0.06	10	200 ± 2	7.3 ± <0.1

Horizon	Data from main site and two subsites (mean values ± standard error)									Data from main site							
	DCB extractable		Oxalate extractable		Fe_o/Fe_d	Fe_d/Fe_t	CEC_{pot} $mmol_c\ kg^{-1}$	BS % of CEC_{pot}	Na saturat. % of CEC_{pot}	Total element concentration							
	Fe	Mn	Fe	Mn						Si_t	Ca_t	Mg_t	Na_t	K_t	Al_t	Fe_t	Mn_t
	mg g^{-1}		mg g^{-1}							mg g^{-1}							
Non-paddy 50																	
Ap	8.27 ± 0.19	0.41 ± <0.01	2.39 ± 0.18	0.35 ± <0.01	0.29	0.22	159 ± 2	>100	2.6	335	22.1	13.5	8.8	20	66	37	0.84
ABw	8.59 ± 0.15	0.42 ± 0.01	2.67 ± 0.21	0.37 ± 0.01	0.31	0.24	153 ± 6	>100	4.1	321	24.5	16.1	10.4	21	77	36	0.80
Bw	8.64 ± 0.08	0.42 ± 0.02	2.57 ± 0.17	0.36 ± 0.01	0.3	0.23	147 ± 2	>100	5.4	321	25.9	15.5	9.4	21	77	37	0.82
BCg	8.98 ± 0.06	0.42 ± 0.01	2.45 ± 0.12	0.35 ± 0.01	0.27	0.24	143 ± <1	>100	7.5	320	26.2	15.7	9.5	21	78	37	0.80
CBg	9.11 ± 0.22	0.46 ± 0.05	2.10 ± 0.15	0.44 ± 0.02	0.23	0.23	134 ± 5	>100	11.8	315	28.6	15.9	9.0	22	79	40	0.97
CBlg	9.58 ± 0.20	0.49 ± 0.01	2.21 ± 0.17	0.44 ± 0.03	0.23	0.25	133 ± 4	>100	14.5	315	28.4	15.8	10.2	22	80	38	0.95
Non-paddy 100																	
Ap1	8.57 ± 0.20	0.40 ± 0.01	1.85 ± 0.08	0.33 ± 0.01	0.22	0.22	185 ± 6	>100	1.2	319	20.4	15.4	9.4	22	81	39	0.87
Ap2	8.69 ± 0.48	0.40 ± 0.01	2.05 ± 0.08	0.30 ± 0.02	0.24	0.23	174 ± 6	>100	1.1	327	19.2	14.5	8.9	21	76	38	0.81
Bw	9.55 ± 0.27	0.43 ± 0.02	1.49 ± 0.02	0.28 ± 0.03	0.16	0.24	170 ± 4	>100	1.3	325	20.4	14.5	8.2	21	76	39	0.89
BCwg1	9.92 ± 0.32	0.43 ± 0.04	1.58 ± 0.14	0.39 ± 0.01	0.16	0.23	172 ± 2	>100	1.9	315	22.4	15.1	8.0	23	81	43	0.97
BCw g2	11.01 ± 0.23	0.59 ± 0.05	1.91 ± 0.20	0.46 ± 0.02	0.17	0.26	170 ± <1	>100	3.1	315	21.8	15.3	8.1	23	82	43	0.94
BCw lg	10.62 ± 0.37	0.56 ± 0.05	2.22 ± 0.20	0.51 ± 0.05	0.21	0.25	176 ± 7	>100	4.7	317	21.4	15.2	8.5	23	80	42	0.91
Non-paddy 300																	
Ap	7.84 ± 0.11	0.41 ± 0.01	1.14 ± 0.03	0.35 ± 0.01	0.15	0.23	159 ± 7	95	0.6	343	10.7	12.3	9.4	20	69	34	0.80
Bw 1	8.45 ± 0.31	0.44 ± 0.03	1.07 ± 0.06	0.35 ± 0.01	0.13	0.24	153 ± 12	>100	0.8	337	11.2	13.8	9.4	20	73	36	0.80
Bw 2	9.38 ± 0.37	0.40 ± 0.02	1.00 ± 0.09	0.33 ± 0.01	0.11	0.23	155 ± 12	>100	1.2	329	12.2	14.5	8.4	21	77	40	0.80
Bwg1	9.53 ± 0.47	0.41 ± 0.04	0.98 ± 0.06	0.37 ± 0.02	0.1	0.24	154 ± 7	>100	1.4	316	22.8	15.8	8.6	22	82	41	0.91
Bwg2	9.94 ± 0.17	0.49 ± 0.03	1.42 ± 0.02	0.39 ± 0.03	0.14	0.23	144 ± 4	>100	2.3	312	26.6	16.1	8.4	23	81	43	0.94
Bwlg	9.30 ± 0.10	0.54 ± 0.03	2.09 ± 0.11	0.48 ± 0.05	0.22	0.23	160 ± 3	>100	4.0	309	26.1	16.6	8.5	23	86	41	0.98
Non-paddy 700																	
Ap1	6.86 ± 0.88	0.42 ± 0.02	2.09 ± 0.74	0.35 ± 0.01	0.3	0.25	183 ± 32	74	0.5	356	9.5	10.5	11.4	17	63	28	0.70
Ap2	6.36 ± 0.44	0.42 ± 0.01	1.82 ± 0.29	0.42 ± 0.01	0.29	0.20	169 ± 19	84	0.7	353	9.5	11.0	9.5	18	64	31	0.79
Bw 1	6.77 ± 0.14	0.47 ± 0.01	1.84 ± 0.23	0.43 ± 0.01	0.27	0.20	170 ± 7	>100	0.7	345	10.5	12.5	10.1	18	68	34	0.82
Bw 2	8.13 ± 0.57	0.45 ± 0.08	1.80 ± 0.35	0.40 ± 0.10	0.22	0.22	166 ± 8	>100	1.0	325	18.2	14.7	9.3	21	79	38	0.67
Bwl1	8.82 ± 0.69	0.51 ± 0.06	1.95 ± 0.27	0.56 ± 0.11	0.22	0.21	168 ± 9	>100	1.5	317	24.6	14.9	8.3	22	80	41	0.99
Bwl2	9.22 ± 0.45	0.55 ± 0.08	2.34 ± 0.35	0.50 ± 0.07	0.25	0.24	156 ± 15	>100	2.2	327	24.7	14.3	9.5	20	72	38	0.98

[a] Calculated from $EC_{1:2.5}$ according to FAO, 2006.
[b] Data from Kalbitz et al. (2013).

Table 2
Soil constituents based on 1400 kg soil mass (1 m^2 * ~ 1 m depth).

	SOM[a] kg m^{-2}	$CaCO_3$ equ.[b] kg m^{-2}	Sand kg m^{-2}	Silt kg m^{-2}	Clay kg m^{-2}	Si_t kg m^{-2}	Ca_t kg m^{-2}	Mg_t kg m^{-2}	Na_t kg m^{-2}	K_t kg m^{-2}	Al_t kg m^{-2}	Fe_t kg m^{-2}	Mn_t kg m^{-2}	CEC_{pot} mol_c kg^{-1}	Fe_d kg m^{-2}	Fe_o kg m^{-2}	$Fe_{o/d}$	Fe_{t-d} kg m^{-2}	Fe_{d-o} kg m^{-2}	Mn_d kg m^{-2}
Mudflat TW0	14.2	58.1	99.6	1087	141	460	45.0	20.7	14.3	27.5	92.7	49.1	1.06	111	10.3	5.6	0.55	38.8	4.7	0.32
Marsh M30	16.7	55.9	68.6	1093	166	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	104	9.5	4.0	0.42	n.d.	5.5	0.29
Paddy 50	18.4	33.5	43.8	910	394	442	35.0	22.3	13.3	31.0	113.3	54.5	1.25	216	13.0	4.1	0.32	41.6	8.9	0.61
Paddy 100	18.2	25.1	42.0	906	409	450	28.1	20.6	12.9	30.4	109.4	56.6	1.41	228	14.4	2.8	0.20	42.1	11.6	0.67
Paddy 300	23.4	10.3	60.0	905	401	455	20.8	20.8	13.0	30.8	110.3	54.8	1.15	229	14.1	2.3	0.17	40.7	11.8	0.52
Paddy 700	18.5	0.0	37.3	793	551	455	8.9	16.5	12.5	31.9	116.8	60.6	0.86	272	18.8	2.6	0.14	41.8	16.2	0.31
Paddy 1000	18.8	0.0	35.0	767	579	453	8.8	16.1	11.4	32.4	118.5	62.7	0.84	322	20.7	2.9	0.14	42.0	17.9	0.34
Paddy 2000	17.3	0.0	58.8	844	480	475	8.8	12.6	11.7	27.3	103.7	58.0	1.03	257	20.5	2.5	0.12	37.5	18.1	0.49
Non-paddy 50	16.2	42.1	52.5	973	316	446	37.8	21.9	13.4	30.0	108.6	53.3	1.25	196	12.7	3.2	0.25	40.5	9.5	0.56
Non-paddy 100	18.3	28.3	32.4	907	414	445	29.7	21.2	11.8	31.2	112.3	58.2	1.27	244	14.2	2.7	0.19	44.0	11.5	0.59
Non-paddy 300	15.9	29.8	48.6	930	376	447	29.4	21.5	12.1	30.7	112.3	56.1	1.26	216	12.9	2.0	0.16	43.2	10.9	0.56
Non-paddy 700	15.4	17.1	53.7	1045	269	461	28.1	19.5	13.1	28.0	103.2	51.8	1.21	232	11.7	2.9	0.24	40.2	8.8	0.66
Average paddies	19.1	11.5	46.1	854	469	455	18.4	18.1	12.5	30.6	112.0	57.9	1.09	254	16.9	2.9	0.18	40.9	14.1	0.49
Average non-paddies	16.4	29.3	46.8	964	344	450	31.2	21.0	12.6	30.0	109.1	54.8	1.25	222	12.9	2.7	0.21	42.0	10.2	0.59

[a] OC * 2 (SOM: soil organic matter).
[b] TIC * 8.3.

the tidal flat (5.0 g kg^{-1}). The highest IC concentration in the uppermost horizon was also found in NP50 with 1.6 g kg^{-1}.

Paddy soils contained less IC than non-paddy soils. P50 was the only paddy soil that contained carbonates throughout the entire profile, with an IC concentration in the uppermost horizon similar to NP50 (1.4 g kg^{-1}). Paddy soils were carbonate-free after 700 years of soil development. The P100 and P300 contained carbonates only in the subsoil, but the values remained below the ones of the non-paddy soils.

Paddy soils lost 191 kg IC ha^{-1} yr^{-1} (159 g $CaCO_3$-equivalent m^{-2} yr^{-1}) by decalcification in the first 300 years of soil development. This decalcification rate was much smaller in non-paddy soils with 113 kg IC ha^{-1} yr^{-1} (94 g $CaCO_3$-equivalent m^{-2} yr^{-1}) in 300 years (Table 2).

3.8. Oxides and total element concentrations

The content of total Fe in the younger soils is 3.5–4%, of which about 20–30% is bound in oxyhydroxides (Tables 1a, 1b). The DCB extractable iron (Fe_d) contents in non-paddy and younger paddy soils were in a narrow range between 6 and 11 g kg^{-1}. In the subsoils of older paddies (≥700 years) Fe_d contents up to 29 g kg^{-1} were observed (Table 1a). In the topsoils Fe_d contents remained in the range between 6 and 9 g kg^{-1}. The ratios of oxalate extractable iron (Fe_o) and Fe_d (Fe_o/Fe_d) in the non-paddy soil ranged between 0.1 and 0.3, without clear differences between top- and subsoil. In the paddy topsoils, the Fe_o/Fe_d ratios increased with time, up to ratios of 0.7. In the subsoil Fe_o/Fe_d ratios decreased systematically with duration of paddy use from 0.3 in P50 to 0.02 in P2000. In non-paddy soils and in P50 the ratio of DCB extractable and total iron (Fe_d/Fe_t) only varied within a narrow range within the whole profile (0.20–0.26). The Fe_d/Fe_t ratio was slightly elevated (0.25–0.28) in P100 and P300, whereas older paddy soils were characterized by high variations of Fe_d/Fe_t within the soil profile (0.22–0.51).

The variability of Mn_d contents in paddy soils (P50–P700) was much higher (0.20–0.78 g kg^{-1}) than in corresponding non-paddy soils (0.40–0.59 g kg^{-1}). The highest amounts of Mn_d could be found in P1000 and P2000 at a depth between 15 and 40 cm (up to 0.9 g kg^{-1}).

The total concentration of basic cations usually increased with increasing soil depth in both the paddy and non-paddy soils, with the highest concentrations of total elements found mainly in the surface soils of the younger profiles (Tables 1a, 1b, 2), which also contained the largest amounts of $CaCO_3$ (see Section 3.7). Changes in Si_t concentrations were small. Highest extractable Si_t concentrations were again found in the surface soils, showing a trend to increased amounts with prolonged paddy or non-paddy management (Tables 1a, 1b).

3.9. pH, electrical conductivity, and cation exchange capacity

The pH (KCl) values remained above 7 as long as carbonates were present. Without carbonates most paddy topsoils were in the range of pH 5–7. Younger paddy subsoils and non-paddy topsoils ranged between pH 6 and 8 (Tables 1a, 1b).

The electrical conductivity ($EC_{1:2.5}$) tended to be lower in older soils but showed different patterns with depth for paddy and non-paddy soils. In the subsoil of NP50 the $EC_{1:2.5}$ values exceeded 650 µS cm^{-1}, falling to values below 200 µS cm^{-1} in the two old non-paddy soils. Non-paddy topsoils show $EC_{1:2.5}$ values between 200 and 300 µS cm^{-1} (NP50 and NP100) and around 150 µS cm^{-1} (NP300 and NP700). The variability of $EC_{1:2.5}$ in the groundwater and irrigation water may explain why $EC_{1:2.5}$ values did not strictly follow a uniform trend along the chronosequences. Based on EC_{SE} values according to FAO (2006), TW0 exceeded the limits for a salic horizon and both lower horizons of NP50 exceeded the limit for a hyposalic qualifier of 4 dS m^{-1} according to IUSS Working Group WRB (2007).

Cation exchange capacity (CEC_{pot}) at pH 7 in the non-paddy soils ranged between 130 and 180 $mmol_c$ kg^{-1} soil, with only slightly higher

Table 3
Microbial biomass C and N (C_{mic}, N_{mic}), as well as total DNA contents measured in 0–20 cm depth in paddy soils under flooded conditions (dw = dry weight). Standard deviations are given in parentheses (n = 5). Significant differences are indicated by different letters.

Soil age	C_{mic}	N_{mic}	DNA content	C_{mic}/N_{mic}
	mg kg^{-1} dw	mg kg^{-1} dw	mg kg^{-1} dw	
TW0	150 (58) *a*	39 (6.8) *a*	140 (35) *a*	3.7 (0.93) *a*
P50	720 (120) *b*	28 (18) *a*	760 (24) *bc*	33 (25) *abcd*
P100	1000 (330) *b*	110 (55) *ab*	630 (120) *b*	9.7 (2.4) *b*
P300	1800 (780) *b*	92 (36) *ab*	810 (220) *bc*	20 (2.6) *c*
P2000	5100 (1300) *c*	150 (30) *b*	1100 (140) *c*	35 (4.1) *d*

values in the humus-rich topsoils. Per kilogram of clay, most horizons showed values between 500 and 850 $mmol_c$ (data not shown). Most values for paddy soils were within the same order. However, the CECs in the uppermost horizons of P700 and P2000 were markedly higher, with >200 $mmol_c$ kg^{-1} soil. In the subsoils of P700, P1000, and P2000, the highest CEC values, up to 280 $mmol_c$ kg^{-1} soil, were found. These subsoils were also richer in clay. The CEC per kilogram of clay did not show changes with depth (data not shown). The highest CEC value was found in a buried A horizon of P1000 with 311 $mmol_c$ kg^{-1} soil.

Base saturation was only meaningful for carbonate-free horizons. In the uppermost horizons of the paddy soils, the base saturation ranged between 70% and 82% for P100 to P1000 but only reached 58% in P2000. The subsoils (P700, P1000, P2000) showed values between 72% and 84%. Base saturation in non-paddy soils tended to be higher (Table 1b).

3.10. Redox regime

Paddy-specific water management creates a typical redox gradient in soil profiles. Independent of the redox conditions prior to flooding (site specific mean of Eh at 10 cm ranged between −12 and +570 mV), all paddy soils followed a similar trend after flooding: topsoils (10 cm) reacted faster than subsoils (35 cm) due to changing water conditions, as flooding occurred from above. After 5 days of flooding, topsoils predominantly turned anoxic: Eh values dropped down to −100 mV at 10 cm and stabilized at a range of −170 (P2000) to −200 mV (P100, P700) after approximately 2 weeks. Subsoils followed this trend with a delay of 5 days (P100, P2000) to 10 days (P700). Differences in the redox regime between the paddy sites after flooding were mainly ascribed to differences in paddy management (planting method, water management). Draining before harvest was accompanied by increasing Eh values: mean Eh values measured 10 days after harvest ranged between −92 (P2000) and +426 mV (P100) at 10 cm and between −142 (P2000) and +350 mV (P100) at 35 cm (individual data not shown).

Such distinct Eh cycles were not observed for non-paddy soils: oxic conditions prevailed throughout the measurement period. Mean Eh values of non-paddy soils ranged from +460 to +600 mV. No significant differences between the sites (NP100, NP700) or depths were found, even though Eh values tended to be slightly lower in topsoil than in subsoil.

3.11. Statistical analyses

To investigate significant differences between individual soil ages, mean values of the selected parameters were compared for each horizon using MANOVA. For both paddy topsoil horizons (Table 4a), short-term changes were identified for IC, and intermediate to long-term changes were detected, e.g. for OC and pH values. In paddy subsoil horizons (Table 4b), intermediate changes were shown for IC and long-term changes were found for pH, and in lower subsoil horizons long-term changes were also found for OC and Fe_d. In non-paddy soils (Table 4c), EC was the only parameter showing significant short-term changes. IC and pH values showed significant intermediate to long-term changes with time of soil development.

A two-step cluster analyses was used to classify the different soil ages into significantly different age groups. Apart from the Ap of non-paddy soils, the cluster analysis assigned all horizons in TW0 and M30 to age group 1 (T1, Tables 4a–4c), accounting for significant differences in most of the investigated parameters ($p < 0.05$). The second age group (T2) revealed different time frames of soil development in paddy topsoils compared to paddy subsoil horizons (Tables 4a, 4b): Of the paddy topsoils, only P50 could be assigned to T2, whereas paddy subsoils of P100 and P300 were also included in the second age group. Consequently, the third soil age group (T3) started with the topsoils of P100,

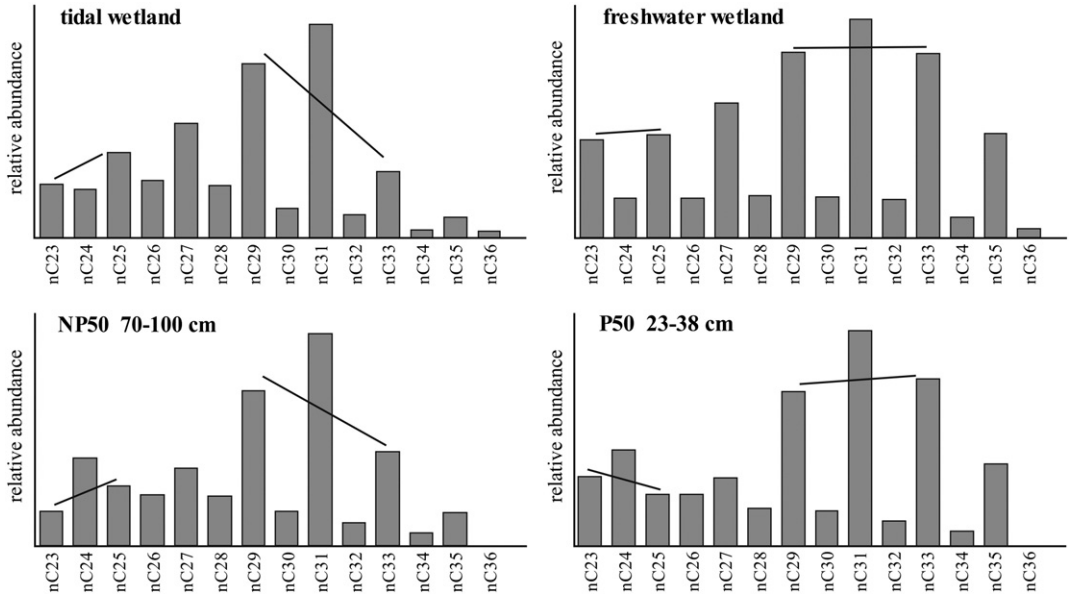


Fig. 3. Distribution of source-diagnostic *n*-alkanes in modern marine and limnic reference sediments compared to those found in paddy and non-paddy subsoils. The lines highlight the proportions of nC_{23} and nC_{25} as well as nC_{29} and nC_{33} which are used to distinguish between marine and limnic substrates. Dominance of nC_{31} alkane in tidal marshes indicates an origin from land plants. In particular grasses derived from riverine input rather than marine biomass. By analogy, the NP50 subsoils developed from a marine substrate, those at the P50 site on a limnic substrate. Most subsoils exhibit a marine tidal flat origin and only a few subsoil horizons at locations close to the coastline before land reclamation show substrate with limnic influence.

but not before 700 years of paddy management in the subsoils. T4, as the oldest soil group, reflected an advanced stage of soil development, which could be found in P2000 (topsoil) and P1000 and P2000 (subsoil). Due to missing age classes of >700 years in non-paddy soils, the cluster analysis separated only two to three age groups (Table 4c). The striking difference between paddy and non-paddy soil development was significantly extended in the second age group (T2), lasting from M30 to NP300 in non-paddy topsoil and from NP50 to NP700 in the lower subsoil horizons. The absence of a third age group in the lower non-paddy subsoil horizons further emphasized the significantly slower soil development within the non-paddy chronosequence.

To evaluate the contribution of each parameter (i.e., variable) to the differentiation between the age groups, structure coefficients were calculated by a discriminant analysis for the evaluated soil horizons (Tables 5a, 5b). IC was one of the most important variables differentiating between age groups 1 and 2 as well as 2 and 3 in paddy topsoils. Only during the first phase in Ardp was BD more important than IC. In paddy subsoils, the contributions of the different variables were more equally distributed. The contribution of IC to differentiating between age groups never exceeded 16%. pH was more important, with values between 22% and 46% to differentiate between T1, T2, and T3. In contrast to paddy topsoils, Fe_d contributed >14% in all age groups of the paddy subsoils.

4. Discussion

4.1. Substrate composition, substrate origin, and land use history

The similarity in soil texture across the whole chronosequence is a strong indicator that all soils developed from similar parent materials. The finer texture of buried horizons of the older paddy soils indicated a preferred sedimentation of finer textured material in former times (closer to the original coastline), whereas slightly coarser material (mainly silt) is currently deposited. The lipid composition of the recent marine tidal flat sediments and the subsoils predominantly reflected an older terrigenous source. The alkyl lipids were dominated by long-chain homologues derived from land plant waxes, whereas short chain homologues indicative of marine algal and bacterial biomass (Schwark, 2009; Volkman et al., 1992) were low in abundance (Fig. 3). This indicated that the production and preservation of marine organic matter in tidal flat sediments was low and overshadowed by the more recalcitrant organic matter derived from the suspended Yangtze River load, which is deposited along the Zhejiang coast by the Eastern China Sea coastal current. Hence, although deposited in a marine sedimentary realm, the organic matter of the tidal flat sediments was mainly terrigenous in origin, in line with earlier reports on allochthonous terrigenic sources of organic matter in tidal flats along the Zhejiang coast (e.g., Bouloubassi et al., 2001; Liu et al., 2008; Zhang et al., 2007).

The *n*-alkanes with intermediate chain length of 23 or 25 C-atoms are derived from aquatic macrophytes and are specifically indicative of freshwater sediments, such as deposits in coastal lagoons and lakes (Ficken et al., 2000). Yet, the limnic sediments studied here likely received local input of alkanes from the surrounding land vegetation (rich in nC_{33} alkane) rather than from the Yangtze hinterland (Fig. 3). Based on these differences, a tidal flat origin was reconstructed for NP50, whereas some subsoil horizons at the site P50 probably developed from lake sediments (Fig. 3). A contribution of limnic substrates to the subsoil horizons of P100, P300, NP300 and P2000 was also in line with the location of these sites along the original coastline (see Fig. 1), i.e., before land reclamation started approximately 2000 years ago.

Following conversion from marine marshland to cultivated paddy or non-paddy soils, fresh wax *n*-alkanes originating from crop plants were introduced into soils and thus modified the inherited lipid pattern. In particular, the relative proportion of nC_{31} originating from cereal crops increases in non-paddy managed fields. This leads to a clear differentiation of the SOC of paddy managed fields (exhibiting an *n*-alkane distribution pattern matching that of tidal flat and limnic substrates) from that of the non-paddy managed soils (Fig. 4). Obviously current management practices have been in use for prolonged times, and the respective *n*-alkane patterns have been preserved in the subsoil (see also Mueller-Niggemann et al., 2012; Schwark, 2009). Usually *n*-alkanes

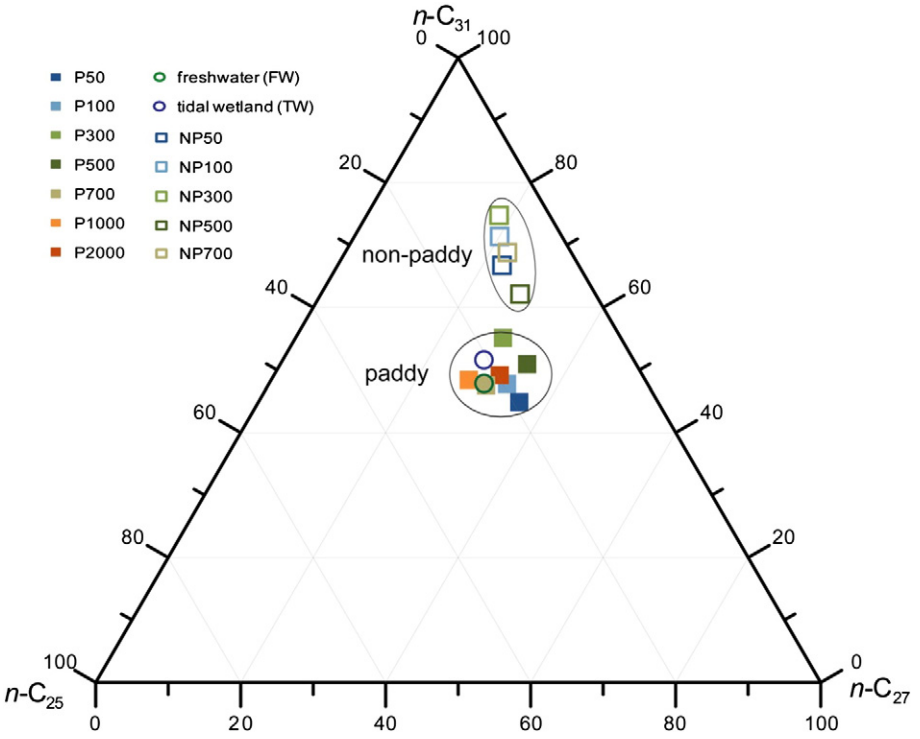


Fig. 4. The *n*-alkane composition of topsoil under paddy versus non-paddy management can be clearly differentiated by decreasing relative proportion of nC_{31} alkanes in paddy soils. Cultivation under submerged conditions generates n-alkane patterns matching those of sub-aquatic deposited limnic and marine substrates.

Table 4a
Mean values of selected soil parameters ordered by horizons (paddy topsoils). Different letters indicate significantly different mean values between soil age within each horizon ($p < 0.05$).

Age group[a]	Soil age	Depth (cm)	OC (mg g^{-1})	IC (mg g^{-1})	BD (g cm^{-3})	$EC_{1:2.5}$ (μS cm^{-1})	pH (KCl)	Fe_d (mg kg^{-1})	Fe_o (mg kg^{-1})
Alp									
T1	TW0[b]	0–30	5.1	5.0	1.31	4180	8.2	7378	4035
	M30[b]	0–13	10.9	4.2	1.31	214	7.8	7994	3394
T2	Paddy 50	0–14	16.4 ab	1.9 b	1.16 ab	670 b	7.4 c	8460 b	4041 ab
T3	Paddy 100	0–15	16.7 ab	0.0 a	1.13 ab	394 ab	5.3 a	8638 b	3136 a
	Paddy 300	0–18	22.6 bc	0.0 a	1.17 ab	399 ab	5.8 ab	8860 b	2875 a
	Paddy 700	0–16	20.9 ab	0.0 a	1.09 ab	546 ab	6.6 bc	8064 b	4417 b
	Paddy 1000	0–16	12.9 a	0.0 a	1.22 b	233 a	5.6 a	8304 b	3655 ab
T4	Paddy 2000	0–15	30.0 c	0.0 a	0.98 a	290 a	4.9 a	5861 a	3383 ab
Ardp									
T1	TW0	0–30	5.1	5.0	1.31	4180	8.2	7378	4035
	M30	13–30	5.3	4.9	1.31	319	8.0	6636	2785
T2	Paddy 50	14–23	5.7 a	2.1 b	1.56 b	379 bc	7.5 c	9995 a	1862 a
	Paddy 100	15–21	6.6 a	0.2 a	1.55 b	303 abc	7.0 bc	10,126 a	1532 a
T3	Paddy 300	18–24	14.9 ab	0.0 a	1.51 b	304 abc	6.6 b	9280 a	2221 a
	Paddy 700	16–22	8.6 a	0.0 a	1.46 b	447 c	6.7 bc	7790 a	2614 ab
	Paddy 1000	16–21	6.1 a	0.0 a	1.50 b	195 a	6.5 b	8839 a	1695 a
T4	Paddy 2000	15–20	20.6 b	0.0 a	1.18 a	188 ab	5.4 a	7205 a	4361 b

[a] Age groups classified according to two-step cluster analysis.
[b] TW0 and M30 were excluded from MANOVA due to missing field replicates.

are not very recalcitrant in soil; however, preservation is possible under more extreme environmental conditions like stagnic water in the paddy soil and/or limited bioaccessibility in the subsoil (Amelung et al., 2008; Schmidt et al., 2011).

4.2. Development of BD

High BD values in soil horizons directly below the plow pan in paddy soils ≥100 years were indicative of a reduction of the tillage depth during the last centuries (second plow pan). This distribution of BD values is typical for paddy soils and the result of repeated puddling with water buffaloes or machines. The achieved soil compaction is desired and required for paddy planting and maintains a strongly reduced permeability even after several decades of paddy management (Janssen and Lennartz, 2006). Accordingly, plow pan formation was observed within the first 50 years of soil development in the soils under study, and it does not show any chronological changes during further pedogenesis.

In our soils, the platy structure, required for the plow pan, was present only in some parts of the layer. We therefore suggest altering the definition of the plow pan in the WRB anthraquic horizon to "a platy structure over at least 25% of its horizontal extension". In our research area, puddling is now done with machines. According to our observations, machine puddling is much shallower and creates a new plow pan above the old one, originating from working by hand or with animals. Expecting shallower anthraquic horizons in the future, we suggest reducing the minimum thickness requirement from 20 to 18 cm (minimum thickness of the CST anthrostagnic epipedon).

4.3. Desalinization, carbonate removal, and acidification

Desalinization is a specific process for soils reclaimed from the sea and generally accompanies land embankment as reflected by a strong decrease in EC_{se} from mudflat to marsh. Soils developing under non-paddy management then show a decelerated, but continuous desalinization, while for paddy soils a nonsystematic enrichment in EC_{se} over marsh soil is likely attributed to salt input from brackish irrigation water. However, most paddy horizons are classified as only slightly (0.75–2 dS m^{-1}) or moderately salty (2–4 dS m^{-1}) according to FAO

Table 4b
Mean values of selected soil parameters ordered by horizons (paddy subsoils). Different letters indicate significantly different mean values between soil age within each horizon ($p < 0.05$).

Age group[a]	Soil age	Depth (cm)	OC (mg g^{-1})	IC (mg g^{-1})	BD (g cm^{-3})	$EC_{1:2.5}$ (μS cm^{-1})	pH (KCl)	Fe_d (mg kg^{-1})	Fe_o (mg kg^{-1})
B1									
T1	TW0[b]	0–30	5.1	5.0	1.31	4180	8.2	7378	4035
	M30[b]	13–30	5.3	4.9	1.31	319	8.0	6636	2785
T2	Paddy 50	23–38	5.1 ab	2.4 b	1.47 a	343 b	7.5 c	9944 a	2289 a
	Paddy 100	21–30	5.8 ab	1.9 ab	1.55 a	323 b	7.4 c	10,302 a	1253 a
	Paddy 300	24–30	7.1 b	0.6 ab	1.61 a	262 b	7.4 bc	9894 a	988 a
T3	Paddy 700	22–45	4.1 ab	0.0 a	1.49 a	318 b	6.7 ab	7707 a	1515 a
T4	Paddy 1000	40–55	2.8 a	0.0 a	1.43 a	233 b	6.7 ab	29,214 b	1761 a
	Paddy 2000	20–27	5.5 ab	0.0 a	1.39 a	107 a	6.0 a	11,810 a	1322 a
B2									
T1	TW0	0–30	5.1	5.0	1.31	4180	8.2	7378	4035
	M30	13–30	5.3	4.9	1.31	319	8.0	6636	2785
T2	Paddy 50	38–50	5.2 b	2.4 b	1.47 ab	326 b	7.5 c	8660 a	2636 b
	Paddy 100	30–50	4.7 b	2.9 b	1.53 ab	323 b	7.4 c	10,768 a	1472 ab
	Paddy 300	30–50	5.3 b	0.5 ab	1.62 b	270 b	7.4 c	10,026 a	895 a
T3	Paddy 700	22–45	4.1 ab	0.0 a	1.49 ab	318 b	6.7 b	7707 a	1515 ab
T4	Paddy 1000	40–55	2.8 a	0.0 a	1.43 a	233 b	6.7 b	29,214 b	1761 ab
	Paddy 2000	35–50	2.6 c	0.0 a	1.53 ab	102 a	6.0 a	14,710 ab	604 a

[a] Age groups classified according to two-step cluster analysis.
[b] TW0 and M30 were excluded from MANOVA due to missing field replicates.

Table 4c
Mean values of selected soil parameters ordered by horizons (non-paddy horizons). Different letters indicate significantly different mean values between soil age within each horizon ($p < 0.05$).

Age group[a]	Soil age	Depth (cm)	OC (mg g^{-1})	IC (mg g^{-1})	BD (g cm^{-3})	$EC_{1:2.5}$ (μS cm^{-1})	pH (KCl)	Fe_d (mg kg^{-1})	Fe_o (mg kg^{-1})
Ap									
T1	TW0[b]	0–30	5.1	5.0	1.31	4180	8.2	7378	4035
T2	M30[b]	0–13	10.9	4.2	1.31	214	7.8	7994	3394
	NP 50	0–17	9.6 a	1.6 b	1.33 a	342 a	7.3 b	8404 ab	2503 a
	NP 100	0–24	9.6 a	0.9 ab	1.34 a	185 a	7.3 b	8623 b	1934 a
	NP 300	0–11	10.5 a	0.1 a	1.32 a	150 a	7.0 b	7844 ab	1136 a
T3	NP 700	0–17	9.7 a	0.0 a	1.27 a	150 a	6.1 a	6698 a	2000 a
B1									
T1	TW0	0–30	5.1	5.0	1.31	4180	8.2	7378	4035
	M30	13–30	5.3	4.9	1.31	319	8.0	6636	2785
T2	NP 50	17–24	6.4 a	3.1 b	1.37 a	417 b	7.5 a	8636 bc	2569 c
	NP 100	24–30	7.0 a	2.1 b	1.44 a	191 a	7.4 a	9548 c	1489 ab
T3	NP 300	11–22	6.6 a	0.1 a	1.39 a	149 a	7.1 a	8451 b	1066 a
	NP 700	17–25	5.3 a	0.3 a	1.38 a	116 a	6.9 a	6768 a	1837 b
B2									
T1	TW0	0–30	5.1	5.0	1.31	4180	8.2	7378	4035
	M30	13–30	5.3	4.9	1.31	319	8.0	6636	2785
T2	NP 50	24–45	5.6 a	3.5 b	1.39 a	459 b	7.5 b	8982 a	2452 b
	NP 100	30–38	6.2 a	2.1 ab	1.49 a	211 ab	7.5 ab	9918 a	1582 ab
	NP 300	32–50	5.3 a	2.7 ab	1.42 a	182 a	7.5 ab	9526 a	982 a
	NP 700	25–45	5.3 a	1.3 a	1.39 a	150 a	7.3 a	8132 a	1798 ab

[a] Age groups classified according to two-step cluster analysis.
[b] TW0 and M30 were excluded from MANOVA due to missing field replicates.

(2006). Rapid desalinization in the study area was also described by Iost et al. (2007), who explained that it occurred by leaching processes due to the high annual precipitation of about 1300 mm.

A Na saturation of ≥15% is expressed by the Sodic qualifier, which again only applied to TW0 where the exchange complex was almost completely saturated with Na. Both the Salic and Sodic qualifiers were no longer applicable 30 years after land reclamation, underlining the fast desalinization after embankment. The Hyposodic qualifier, which needs a Na saturation of ≥6%, was present in the marshland, NP50, and the paddy soils up to P700.

Paddy management resulted in enhanced decalcification of the marine sediments compared to the non-paddy agricultural use, typically with a faster removal of calcite than dolomite (Fig. 2). Whereas the carbonate loss in the paddy topsoil was much faster compared to non-paddy topsoils, decalcification and related processes in the subsoil were only slightly accelerated. A detailed discussion about the relevance of carbonate weathering due to paddy cultivation is given in Kalbitz et al. (2013).

The decalcification rate of the investigated non-paddy sites was similar to a study of Mueller (1994), who described a complete loss of carbonates only in the upper 50 cm of the soil profile after 700 years of soil development in a diked, silty-clayey marshland soil in northern Germany, where parent marine sediments had a similar carbonate content compared with the sediments of the Hangzhou Bay. Iost et al. (2007) also observed an incipient loss of carbonates in 20-year-old reclaimed marsh topsoils from Zhejiang Province, P.R. China. The Calcaric qualifier indicates the presence of ≥2% carbonates in a depth range of 20–50 cm. Whereas the non-paddy soils were calcaric until NP300, P100 was the oldest soil with the Calcaric qualifier in the paddy sequence. Chen et al. (2011) considered carbonate loss as a completely soil intrinsic threshold, not resulting from anthropogenic activities. In contrast, our results now clearly show that specific paddy management accelerates carbonate removal from the paddy soils compared to non-paddy soils.

Several aspects of the specific management of paddy soils are considered responsible for the accelerated loss of carbonates in the topsoils. In contrast to non-paddy soils, paddy management is characterized by inundation resulting in larger water fluxes in paddy soils. These water fluxes are largely lateral, with only limited downward fluxes, mainly in preferential flow paths if the plow pan is well established (Janssen and Lennartz, 2006). Intensive decalcification is generally observed in hydric soils subject to periodic changes between oxidation and reduction, associated with increased carbon dioxide pressure in the soil during periods of waterlogging (van den Berg and Loch, 2000). The increased accumulation of organic matter and thereby ammonium in the water layer under paddy management may then in turn promote acidification due to higher acid production during the dry cropping phase, like a positive feedback loop between a seasonally decoupled SOC accumulation and SOC oxidation. A larger input of nitrogen fertilizer to paddy soils during the last ca. 50 years may additionally contribute to a more intensive production of protons after ammonia oxidation and dissolution of carbonates (Barnes and Raymond, 2009; Gandois et al., 2011; Perrin et al., 2008).

The pH values (KCl) of the calcareous horizons were between 7 and 8. After decalcification a rapid acidification occurred, leading to values of pH 6 in non-paddy topsoils, of pH 5 in paddy topsoils, and to values slightly above pH 6 in the subsoils of the paddy soils. With increasing age, a clear loss of Ca_t and Mg_t was observed (Table 2), being again stronger in paddy soils (P700 about 80%) than in non-paddy soils (NP700 about 40%) due to the weathering of calcite and dolomite. In contrast, the stocks of Na_t and K_t did not decline systematically for either paddy or non-paddy soils (Table 2). When taking into account that K and Na salts are usually more soluble than their Ca and Mg counterparts, a current replacement of these former ions with irrigation water and fertilizers, respectively, may have contributed to this finding.

In summary, the losses of salt and carbonates proceeded in different time frames: the desalinization of the total soil profile was almost finished 50 years after embankment, while the leaching of the carbonates and related acidification required several hundred years, i.e., they covered different phases of soil development. These processes were clearly accelerated in paddy sites compared to non-paddy sites. Paddy soils were characterized by a rapid decalcification of the puddled horizons (decalcification rate of approx. 50 g kg^{-1} yr^{-1} vs. approx. 40 g kg^{-1} yr^{-1} in non-paddy topsoils during 100 years of soil development). It was complete 100 years after embankment, whereas carbonate removal below the plow pan progressed at a rate that was only slightly higher than in the non-paddy soils (approx. 31 g kg^{-1} yr^{-1} in paddy subsoils, and 29 g kg^{-1} yr^{-1} in non-paddy subsoils during the first 100 years of soil development [B1 horizon, Tables 5a, 5b]).

Table 5a
Structure coefficients derived from discriminant analysis and their contribution to differentiate between the four age groups in Alp, Ardp, B1 and B2 horizons.

	Variable	Phase I (T1 ⇔ T2)		Phase II (T2 ⇔ T3)		Phase III (T3 ⇔ T4)	
		Structure coefficient	Contribution (%)	Structure coefficient	Contribution (%)	Structure coefficient	Contribution (%)
Alp	IC	0.1 55	60.5	0.863	93.1	–[a]	–[a]
	pH	–[a]	–[a]	0.029	3.2	0.111	12.9
	EC	–[a]	–[a]	0.021	2.2	0.049	5.7
	OC	−0.065	25.2	−0.004	0.9	−0.182	21.1
	BD	0.037	14.3	0.001	0.5	0.177	20.5
	Fe_o	–[a]	–[a]	0.008	0.1	0.015	1.7
	Fe_d	–[a]	–[a]	0.000	0.0	0.328	38.1
	Absolute value	0.257	100	0.926	100	0.861	100
	Wilks' Lambda	0.001 (p = 0.014)		0.001 (p = 0.000)		0.034 (p = 0.000)	
Ardp	IC	−0.175	19.7	0.784	37.6	−0.036	1.6
	pH	–[a]	–[a]	0.589	28.2	−0.526	24.0
	EC	–[a]	–[a]	0.144	6.9	−0.154	7.0
	OC	0.157	17.8	−0.163	7.8	0.297	13.6
	BD	0.554	62.5	0.194	9.3	−0.608	27.7
	Fe_o	–[a]	–[a]	−0.045	2.1	0.402	18.4
	Fe_d	–[a]	–[a]	0.169	8.1	−0.169	7.7
	Absolute value	0.886	100	2.088	100	2.191	100
	Wilks' Lambda	0.011 (p = 0.077)[b]		0.184 (p = 0.025)		0.089 (p = 0.005)	
B1	IC	−0.120	16.2	−0.085	14.0	–[a]	–[a]
	pH	−0.163	22.0	–0.281	46.4	−0.100	13.7
	EC	−0.082	11.0	0.009	1.6	−0.283	39.0
	OC	0.021	2.9	−0.068	11.2	−0.031	4.2
	BD	0.137	18.6	−0.042	7.0	−0.052	7.2
	Fe_o	−0.103	13.9	0.000	0.1	–[a]	–[a]
	Fe_d	0.114	15.4	−0.119	19.7	0.261	35.9
	Absolute value	0.740	100	0.606	100	0.727	100
	Wilks' Lambda	0.008 (p = 0.000)		0.012 (p = 0.000)		0.066 (p = 0.090)[b]	
B2	IC	−0.175	15.3	0.180	15.5	–[a]	–[a]
	pH	−0.265	23.2	0.495	42.4	−0.037	10.5
	EC	−0.144	12.6	−0.030	2.6	−0.093	26.1
	OC	−0.031	2.7	0.177	15.2	−0.100	28.1
	BD	0.219	19.2	0.070	6.0	−0.028	7.8
	Fe_o	−0.143	12.5	0.021	1.8	−0.016	4.4
	Fe_d	0.166	14.5	0.193	16.5	0.082	23.1
	Absolute value	1.143	100	1.166	100	0.357	100
	Wilks' Lambda	0.024 (p = 0.005)		0.046 (p = 0.005)		0.008 (p = 0.023)	

[a] Not processed by discriminant analysis.
[b] Differentiation between age groups statistically not significant (α = 0.05).

4.4. Weathering and formation of pedogenic minerals

There was no apparent trend for gains or losses either with age of the soils or between paddy and non-paddy soils for quartz, feldspars, mica, and the clay minerals kaolinite, chlorite, and vermiculite (Fig. 2). The slight differences were most probably due to the slight variability of the parent material. This was supported by the total contents of elements (Si_t, K_t; Tables 1a, 1b).

The time for silicate weathering after decalcification of the quartz-rich parent material in an environment with high ion concentrations in the soil solution was obviously too short to cause significant differences in the assemblage of silicates. This was in strong contrast to the finding of Li et al. (2003) who observed a significant loss of K-bearing clay minerals and an increased formation of less siliceous, more ferromagnesian, chlorite over an 80-year cultivation period in soils developed from Quaternary red clays in Jiangxi Province. These differences may be explained by the fact that the substrate, tidal flat sediments, derive from eroded soil, i.e., they had already undergone substantial weathering and diagenesis before deposition and paddy soil formation.

Major differences were found between the stocks and profile distribution of Fe and Mn oxides between paddy and non-paddy soils. On a whole soil basis, the stocks of Fe_o decreased from young to old in paddy and non-paddy soils more or less in the same range (Table 2), but crystalline Fe oxides (Fe_{d-o}) increased in paddy soils much more than in non-paddy soils. Since paddy soils have on average only 1.0 kg m^{-2} less Fe_{t-d} than non-paddy soils but 4.0 kg m^{-2} (Table 2) more Fe_d, we concluded that an enrichment of Fe either with the irrigation water and its suspended mineral load, with the groundwater, or with both caused the Fe_d enrichment rather than an increased Fe-(hydr)oxide formation by silicate weathering. The more efficient water storage in paddy than in non-paddy soils thus appears to have affected both nutrient replacement (Na, K) and some redox-sensitive element re-additions (Fe); however, we did not find any evidence of a re-introduction of Mn. On a whole profile basis (Table 2), the stocks of Mn_t and Mn_d decreased from young to old paddy soils in contrast to non-paddy soils, where no trend was observable. The variability of Mn_d contents in paddy soils was much higher than for Fe_d, due to the stronger sensitivity of Mn to redox cycles.

Long-term paddy use, however, affected the re-distribution of elements in the profile. The old paddy topsoils exhibited a loss of Fe_d, which was accompanied by a gain in the subsoil (Table 1a). We concluded that topsoil Fe_d was almost completely reduced during repeated flooding, partially transported into the subsoil, and partially re-oxidized in the topsoil in a less crystallized form. In addition, the higher OC concentration in older paddy topsoils may have had an inhibitory effect on Fe oxide crystallization (Schwertmann, 1966). The increase of Fe_d in the subsoil was caused mainly by crystalline Fe (Fe_{d-o}). This

Table 5b
Structure coefficients derived from discriminant analysis and their contribution to differentiate between the three age groups in Ap, B1 and B2 horizons.

	Variable	Phase I (T1 ⇔ T2)		Phase II (T2 ⇔ T3)	
		Structure coefficient	Contribution (%)	Structure coefficient	Contribution (%)
Ap	IC	0.067	**5.6**	0.118	**10.8**
	pH	0.075	**6.2**	0.439	**40.0**
	EC	0.853	**70.9**	0.088	**8.0**
	OC	−0.136	**11.4**	0.021	**1.9**
	BD	−0.020	**1.7**	0.158	**14.4**
	Fe_o	–[a]	–[a]	0.002	**0.1**
	Fe_d	−0.051	**4.2**	0.272	**24.8**
	Absolute value	1.202	**100**	1.097	**100**
	Wilks' Lambda	0.005 (p = 0.000)		0.056 (p = 0.003)	
B1	IC	0.159	**17.4**	0.499	**36.8**
	pH	0.301	**32.9**	0.247	**18.2**
	EC	0.077	**8.4**	0.169	**12.5**
	OC	−0.130	**14.2**	0.079	**5.8**
	BD	−0.088	**9.6**	0.051	**3.8**
	Fe_o	–[a]	–[a]	0.112	**8.3**
	Fe_d	−0.161	**17.5**	0.198	**14.6**
	Absolute value	0.915	**100**	1.355	**100**
	Wilks' Lambda	0.008 (p = 0.024)		0.038 (p = 0.003)	
B2	IC	−0.318	**15.2**		
	pH	−0.584	**27.8**		
	EC	−0.292	**13.9**		
	OC	0.085	**4.1**		
	BD	0.231	**11.0**		
	Fe_o	−0.308	**14.7**		
	Fe_d	0.279	**13.3**		
	Absolute value	2.098	**100**	0.606	**100**
	Wilks' Lambda	0.088 (p = 0.004)		0.012 (p = 0.000)	

[a] Not processed by discriminant analysis.

observation was in line with that of Thompson et al. (2006), who described an increasing crystallinity of Fe oxides during soil redox alternation (200–700 mV) in short-term batch experiments (56 days).

In summary, our results point to redistribution and transformation of oxides as Fe_d and $Fe_{d/t}$.

The formation of the hydragric horizon is explained by a slow but constant movement of reduced Fe and Mn from the puddled layer through the plow pan to the subsoil (Driessen et al., 2001; Gong, 1983). If reducing conditions predominate in the macropores of the hydragric horizon (due to the mineralization of OM), Fe and Mn are further transported into the interiors of the aggregates. The intra-aggregate pore system may still contain some oxygen, especially because there is nearly no organic matter in the interiors of the aggregates and therefore the oxygen has not been used up to oxidize OC. Thus, Fe and Mn are preferentially reoxidized in the interiors of aggregates, leading to a stagnic color pattern. In fact, the paddy subsoils showed markedly higher Fe_d concentrations with increasing age, and the hydragric horizons were characterized by a stagnic color pattern. Eventually, Fe and Mn were even leached out from the upper part of the hydragric horizon leaving it as an albic horizon. Because stagnic characteristics and albic horizons are inherent to hydragric horizons, there is no need to add the respective qualifiers.

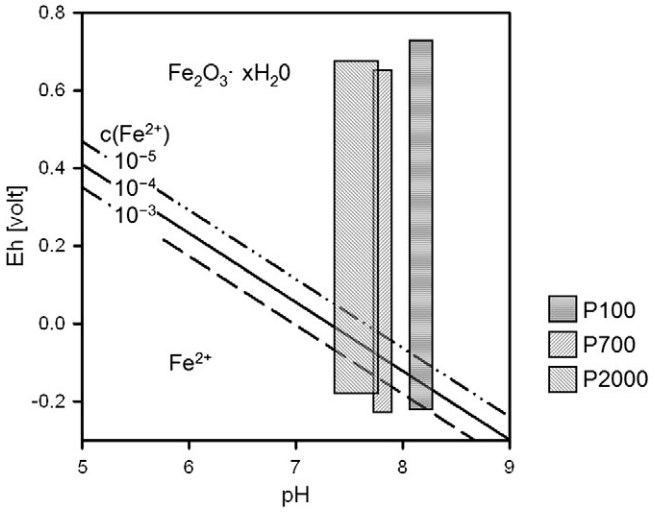


Fig. 5. Eh–pH–stability diagram: Stability area of ferrihydrite ($Fe_2O_3 \cdot xH_2O$) relative to Eh, pH and an Fe^{2+} activity = 10^{-5}, 10^{-4} and 10^{-3} mol l^{-1}. Dashed squares illustrate Eh–pH-ranges as measured within the subsoils of the different paddy sites.

Considering these processes, the definition of the hydragric horizon in WRB requires one or more of the following: first, Fe or Mn coatings or concretions; second, higher dithionite–citrate extractable Fe (≥2 times) or Mn (≥4 times) compared with the surface horizon; or third, redox-depleted zones in macropores. However, it is unlikely that these three features will occur together in the entire hydragric horizon. Therefore, we suggest a change in the definition from the wording "having one or more of the following:" to "having in every subhorizon one or more of the following:". This would also bring the WRB hydragric horizon closer to the hydragric horizon of CST in which the vertical differences within the horizon are specified. Additionally, the first criterion is rather vague and satisfied by any redoximorphic color pattern. Even a color pattern caused by rising groundwater, which has nothing to do with paddy soil formation, fulfills the criterion. We suggest modifying the definition by: "Fe or Mn mottles inside the aggregates covering 15% or more of the soil volume".

4.5. Accumulation of organic matter

Paddy soil management has a clear effect on the accumulation of soil organic matter. Organic C concentrations of the topsoils increased continuously during 2000 years of paddy use, whereas for a period of 700 years they remained constant in the non-paddy soils. Investigations

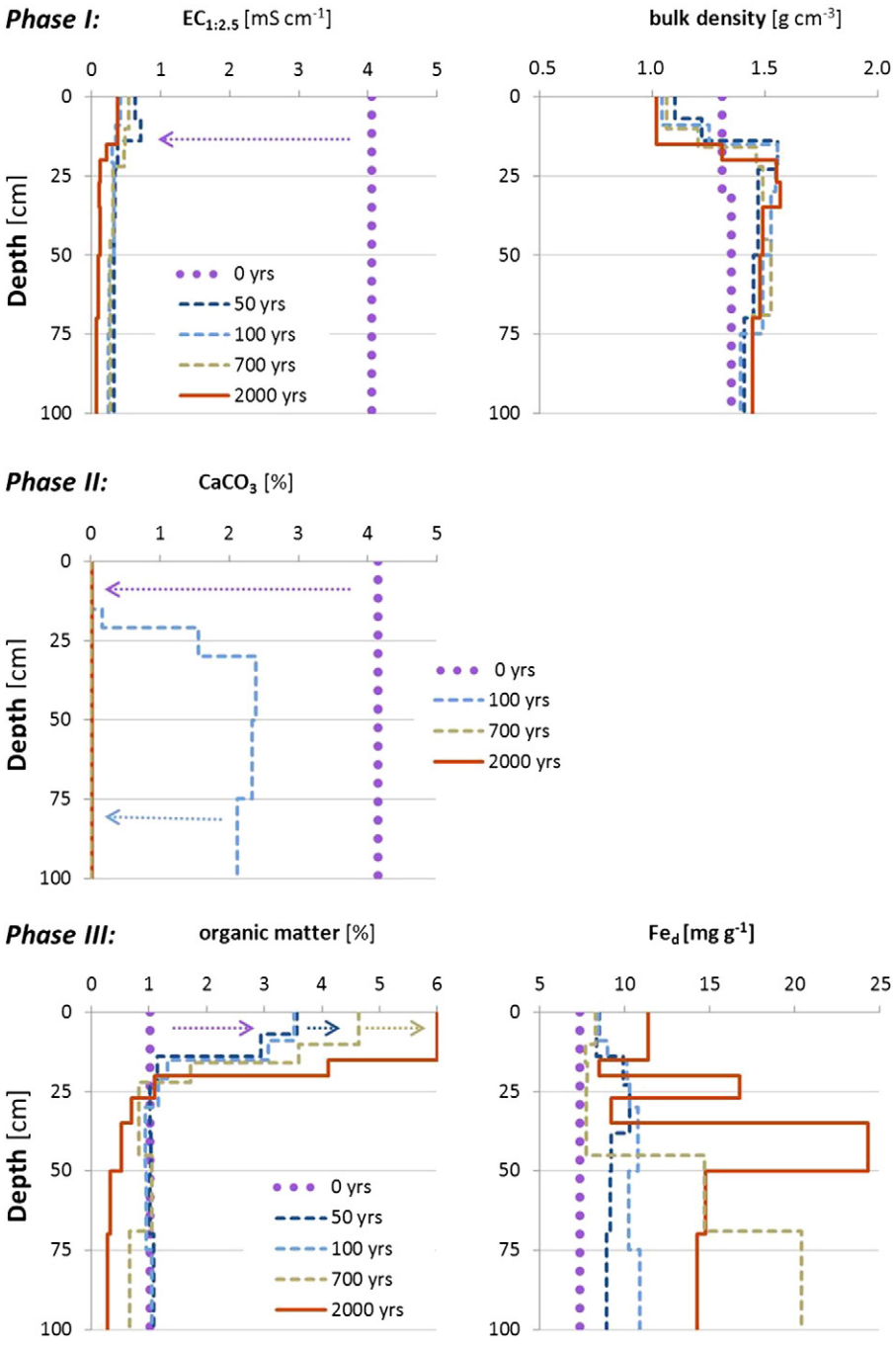


Fig. 6. The indicative processes of the three phases of paddy soil formation. *Phase I*: desalinization (EC values) and development of bulk density distribution. *Phase II*: decalcification. *Phase III*: organic matter accumulation, Fe_d redistribution.

on OC saturation of particle size fractions confirmed the ongoing OC accumulation throughout 2000 years of paddy soil development (Wissing et al., 2011). Positive effects of paddy management on topsoil SOC contents were already described by Cheng et al. (2009) and Sahrawat (2004). Our results were also in accordance with Liu et al. (2006), Shang et al. (2011) and Wu (2011), whose data indicated higher OC concentrations and stocks in paddy soils compared to other arable ecosystems in China. The remarkably small OC concentrations in the topsoil of P1000 can be related to dike maintenance where topsoil material from the paddy field was used (Wissing et al., 2011; Cao, pers. communication). This is confirmed by Wissing et al. (2011), who found a low OC enrichment in the clay-sized fraction of this soil, showing similar values to the 50-year-old paddy soil.

The formation of a dense plow pan leads to a specific pattern of OM accumulation in paddy soil profiles, differentiating topsoil and subsoil. As leaching of DOC to the subsoil is much decreased in paddy soils, OC accumulation is mainly confined to the topsoil, and at the same time OC input to subsoils by roots may be reduced. This results in steep gradients in SOC concentrations in the soil profile in paddy compared to non-paddy soils, and lower accumulation of microbial residues (Roth et al., 2011). In addition, the subsoil horizons of paddy soils lost inherited, marine-derived OC during 2000 years of paddy soil development. This was particularly evident after 700 years of soil development with decreasing OC concentrations in the paddy subsoils, while N_t concentrations are still in the range of TW0 and M30. The selective loss of OC is reflected by narrow C/N ratios in the respective paddy subsoils (P700–P2000). The

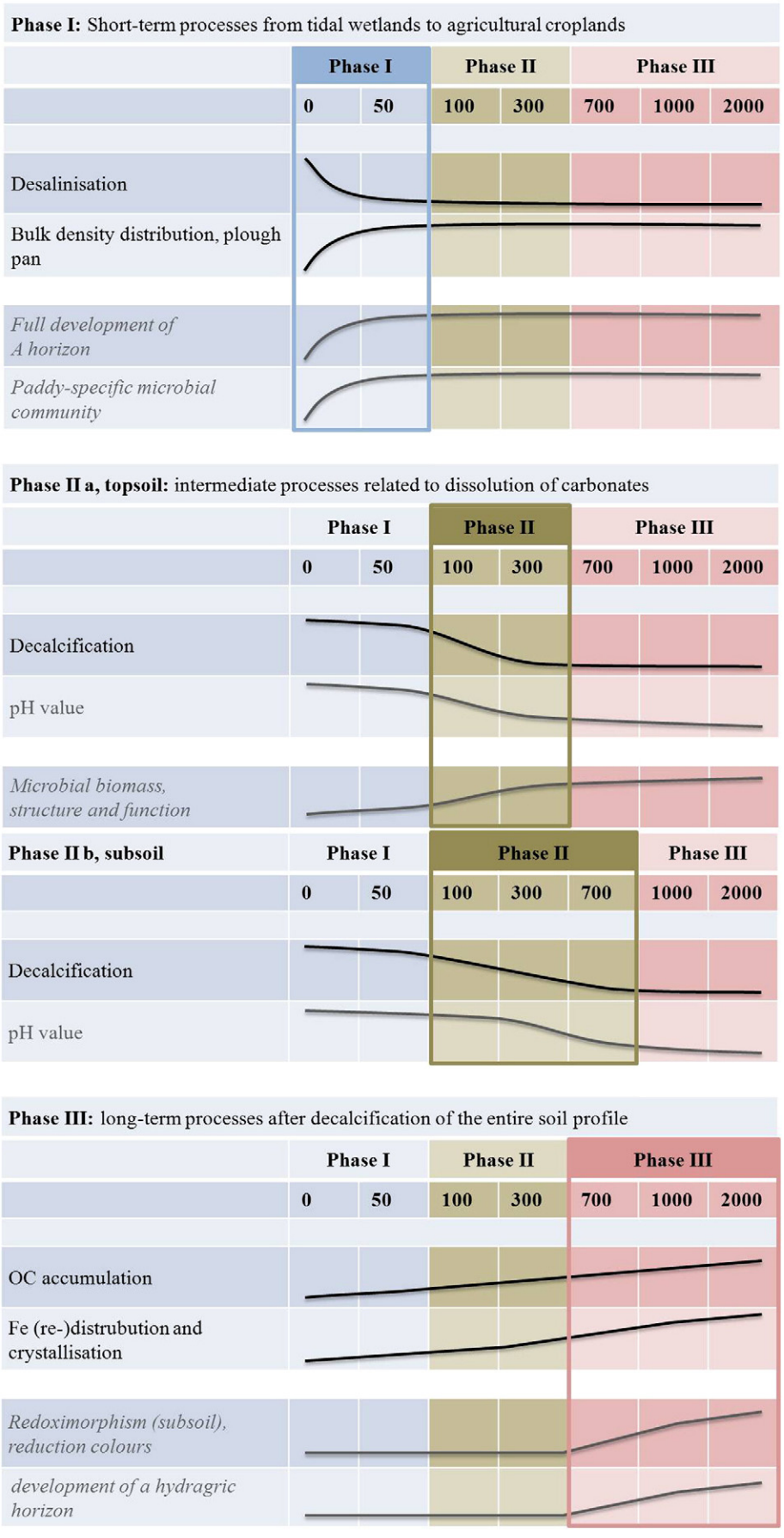


Fig. 7. Three phases of paddy soil formation: a schematic overview.

study of Roth et al. (2011) investigated the proportion of mineral N to total N. The authors showed that in the top 20 cm of paddy soils, the sum of nitrogen found in nitrate and ammonium did not exceed 1.5% of total nitrogen, whereas in the subsoils mineral N fractions contributed up to 4% to total N. Fixed ammonium could also occur in flooded rice soils (Nieder et al., 2011), since the clay fraction of the chronosequence soils consist of illite, kaolinite, chlorite and secondary chlorite, with illite being the major clay mineral (Wissing et al., 2013).

Generally, the accumulation of N_t was closely related to OC during paddy soil development. The narrow C/N ratios indicated an accumulation of fairly decomposed organic matter in the paddy soils of the chronosequence, even if the total amount of N stored in microbial residues was low (Roth et al., 2011).

To summarize, SOC accumulation in paddy topsoils is a long-term process, which seems to proceed beyond 2000 years of paddy soil formation. It might be the driver for the increasing C_{mic} and N_{mic} during paddy soil development (Anderson and Domsch, 1989; Powlson et al., 1987), which reached a maximum only in the 2000 years cultivated paddy soil (Table 3). The disproportional increase of C_{mic} and to lower extent also N_{mic} compared to the amount of extracted DNA can be taken as a measure for changes in the microbial community structure and the related ecophysiology of the organisms. Obviously microbes harboring soils at later stages of paddy soil evolution can store larger amounts mainly of carbon compared to microbes living in TW0 and P50. However, the increasing ratio of C_{mic}/N_{mic} might indicate nitrogen limitation in paddy soils that were cultivated for a longer time period (Ottow, 2011). This is in line with the observed reduced relative number of denitrifiers in soils from P2000 compared to P300 and P50, indicating a lower availability of nitrate in soils, with longer paddy soil history (Bannert et al., 2011).

4.6. Implications for soil classification

Soil formation and the differences between paddy and non-paddy soils were generally well reflected by WRB classification. The original Fluvisols rapidly transformed into Cambisols, and only the paddy soils further developed into Anthrosols. However, some redefinitions concerning the classification of Anthrosols are suggested:

- for the anthraquic horizon and with respect to the plow pan, we suggest to reduce the requirement of a platy structure to a presence in 25% or more of its volume. With respect to the entire anthraquic horizon, we suggest reducing the minimum thickness requirement from 20 to 18 cm.
- for the hydragric horizon, we suggest allowing a subdivision of the hydragric horizon into several subhorizons to consider that different redoximorphic features may occur in different soil depths. We also recommend a more precise definition concerning the accumulation of Fe or Mn in coatings or concretions, requiring Fe or Mn mottles inside the aggregates covering 15% or more of the soil volume.

4.7. Time phases of paddy soil development

Our concept of paddy soil formation was based on a detailed biogeochemical analysis of a soil chronosequence with a rigorous establishment of similar composition of the parent material as well as long-term paddy management based on biomarker studies. The major changes in the characteristic paddy soil properties are exemplified for the three phases observed in the present study in Fig. 6; a schematic overview is given in Fig. 7.

In non-paddy topsoils, EC was the most important variable for distinguishing between age groups 1 and 2, whereas decalcification-related variables (pH and IC) became more important in subsoils and for separating between age groups 2 and 3 in non-paddy topsoils (Table 5b). OC, BD, and Fe_o did not play an important role in differentiating between non-paddy age groups. This underlines the massive loss of carbonates, exchangeable bases, and soluble salts during soil formation, affecting the total soil profile. The specific pattern of OC accumulation in paddy soils with accumulation of OC in the puddled layer and OC loss in the subsoil was reflected by a positive correlation coefficient between soil age and OC in the paddy topsoil ($r = 0.53$, $p \leq 0.05$) and a negative correlation in the lower paddy subsoil horizon ($r = -0.74$, $p \leq 0.01$). The data set for the non-paddy chronosequence clearly reflected a low degree of soil development, with predominating decalcification-related processes. As a consequence, only two early phases of soil formation could be distinguished. In contrast, the data set for the paddy chronosequence allowed the differentiation of three consecutive phases of soil formation associated with specific interdependent processes (Table 5a).

During the initial phase (from T1 to T2; 0–50 years), most of the fundamental changes of soil properties were strongly related to the transition from tidal wetlands to agricultural croplands. This included short-term processes like desalinization of the entire soil profile (Figs. 6, 7), which is a typical feature for arable land reclaimed from the sea. Already at this stage the establishment of a specific microbial community was occurring. Significant changes in diversity and the abundance pattern of functional genes of nitrogen fixation, nitrification, and denitrification were observed between the tidal wetland and the 50-year cultivated paddy soil (Bannert et al., 2011). The development of a poorly permeable plow layer was also completed during the first decades of paddy soil development. At the least, this leads to partly decoupled soil development in topsoil and subsoil horizons. During this first phase the specific features of paddy soil formation were restricted to the formation of the topsoil anthraquic horizon, associated with the development from Tidalic Fluvisols (Calcaric) to Anthraquic Cambisols (Calcaric). No paddy-specific chemical processes were induced at this stage in the subsoil, as indicated by the redox measurements in association with the high pH of the still carbonaceous subsoils. In the next phase of paddy soil development (T2 to T3; 50–100 years in topsoils, 50–700 years in subsoils), the most important process was the dissolution of carbonates. Due to the frequent flooding and drainage, this process was significantly accelerated as compared to soil development under non-paddy agricultural management under the same climatic conditions and leads to Anthraquic Cambisols without the Calcaric qualifier. Within the paddy soil profile, we could see a still higher intensity of paddy-specific processes in the topsoil compared to the subsoil (Figs. 6, 7). Whereas the decalcification process in the regularly flooded topsoil was much faster compared to non-paddy topsoils, decalcification and related processes in the subsoil were only slightly accelerated due to the decoupling impact of the plow pan.

The decalcification of the entire soil profile allowed long-term paddy-specific processes to start in the third phase (T3 to T4; 700 to >2000 years). Mn and Fe, which were reduced in the topsoil, passed through the plow layer and accumulated in the subsoil, where they crystallized in the interiors of the aggregates and served as the origin for the hydragric horizon. These soils are classified as Hydragric Anthrosols. Redox measurements demonstrated that redox-induced dissolution of Fe hydrous oxides and, hence, a release of Fe^{2+} occurred only when lower pH values were reached after complete decalcification (Fig. 5); whereas Eh–pH values of the "younger" paddy site (P100) were in the range of the stability zone of ferrihydrite ($Fe_2O_3 \cdot xH_2O$), the Eh–pH values at the older paddy sites (P700 and P2000) partly extend to the Fe^{2+} stability zone (Fig. 5). This facilitated an on-going differentiation of the Fe_d depth distribution as shown in Fig. 6. OC accumulation was restricted to paddy topsoils and was observed throughout the chronosequence, but decelerated with time (Fig. 6). It was a possible driver for the significantly increasing C_{mic} with cultivation time in the paddy topsoils, measured under flooded conditions (Bannert et al., 2011).

Our dataset allows a more comprehensive and refined concept of paddy soil development than previous ones. Gong (1983) divided paddy soils into three stages of leaching: weakly leached (loss of soluble

salts and carbonates—corresponding to our phases 1 and 2), moderately leached (loss of exchangeable bases), and intensively leached (loss of Fe). Similar results were obtained by Chen et al. (2011). But leaching of ions is only one aspect of paddy soil formation. Our study provides advanced information on microbial biomass and OC enrichment, oxide formation, and leaching and redistribution processes, which are necessary to understand interacting processes of soil formation. We are now able to describe the first stages of soil development in more detail and to identify the early development of a plow pan as a decisive process. Our results clearly indicate that it is necessary to differentiate the rates of soil formation for topsoil- and subsoil-related processes. The hydragric horizons in our soils had higher clay concentrations at some depth. The migration of clay postulated by Chen et al. (2011) to occur in 300- to 700-year-old paddy soils in our chronosequence, delineated from clay content differences, was not confirmed by detailed micromorphological analyses. The micromorphological analyses of subsoil clearly indicated that clay illuviation did not take place in any paddy soil, even the oldest. Therefore, we attributed the higher clay contents to lithological discontinuities, in agreement with the detection of buried A horizons in P700, P1000, and P2000. The high clay contents may have been due either to local additions of lacustrine or fluviatile sediments or to varying sedimentation conditions of the marine sediments. The translocation of clay was most probably isochronic with the weathering of silicates and the formation of pedogenic clay minerals. These processes are associated with a lower pH. Obviously, they were not taking place within the 2000 years that were covered by our chronosequence. However, it was postulated that they occur if suitable soil conditions are reached, especially with respect to pH in a distinct fourth phase of paddy soil formation.

5. Conclusions

The present study investigated a chronosequence of soils developed on mainly marine, calcareous substrates under similar climatic conditions with well-known starting dates of terrestrial soil formation. Agricultural land use began immediately after desalinization. The comparison of two cropping systems—one with paddy rice and one with non-flooded crops—provided a unique opportunity to document the effect of paddy soil management and the associated redox cycling over long time periods on the formation of soil minerals and soil organic matter properties and their interactions.

The dynamics of paddy soil development in direct comparison to soils used for non-flooded crop production were strongly accelerated and led to faster soil formation under wetland rice production. By integrating all identified relevant processes, three phases of paddy soil formation were designated: (1) short-term processes like desalinization (specific for marine sediments) and development of a compacted plow pan form the basis for the next phase (2), which is mainly characterized by decalcification, differentiated in topsoil and subsoil time domains, and (3) long-term processes like SOC accumulation, but also (trans-) formation of oxides, which occurred to a considerable extent after complete decalcification of the entire soil profile. But 2000 years of paddy soil formation (>1300 years of weathering after decalcification) on these terrigenous sediments was too short for a fourth phase of paddy soil formation, which would be characterized by silicate weathering and clay migration.

Supplementary data to this article can be found online at http://dx.doi.org/10.1016/j.geoderma.2013.09.005.

Acknowledgments

The authors thank Rui Yin and the Institute of Soil Science, Chinese Academy of Sciences, in Nanjing for support during sampling at the chronosequence site around Cixi and for logistic handling. We thank Monika Heilmeier for technical assistance and Martina Bauer, Carolin Botond, Robert U. Hagemann, Tahereh Javaheri, Juliane Teichmann, and Maria Vonach for student research assistance. We would also like to thank both anonymous reviewers for their suggestions. The detailed comments were very helpful to improve the manuscript.

We acknowledge the staff of "Boldface Editors" (International Scientific Editorial Service) for editing and proof reading the manuscript. We are grateful to the Deutsche Forschungsgemeinschaft (DFG) for their generous funding of Research Unit FOR 995 "Biogeochemistry of paddy soil evolution."

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